WSQ COURSE SLIDES

# Generative AI for 3D Modeling

## Course Code: TGS-2023037544

Tertiary Infotech Academy Pte Ltd
UEN 201200696W

Trainer: _____

**Version v8.0 · 15 August 2026**



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COURSE ADMINISTRATION

00

# Welcome, outcomes and assessment

Four days · 32 hours · Blender-based delivery


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# Digital Attendance (Mandatory)

1

## AM

Complete the SSG digital attendance at the start of each day.

2

## PM

Complete the afternoon attendance after lunch.

3

## Assessment

Complete the assessment attendance before the WA and PP.

4

## Funding

At least 75% attendance and a Competent result are required.


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# About the Trainer




## General Trainer

Complete this profile before class

**NAME** _____

**TITLE / DESIGNATION** _____

**QUALIFICATIONS** _____

**AREAS OF EXPERTISE** _____

**TRAINING AND INDUSTRY EXPERIENCE** _____

**CONTACT** _____


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# About the Trainer




## Dr Alfred Ang

Principal Trainer · Tertiary Infotech Academy

| | |
|---|---|
| ROLE | Principal Trainer and courseware lead |
| CERTIFICATION | Adult education and applied technology training |
| DELIVERS | AI, automation, data and digital design programmes |
| FOUNDER | Tertiary Infotech / Tertiary Courses |
| APPROACH | Concept-led teaching with evidence-based practice |
| CONTACT | enquiry@tertiaryinfotech.com |


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# Let’s know each other

1

## You

Name, role and organisation.

2

## Experience

Your exposure to Blender, 3D design or generative AI.

3

## Use case

A model or workflow you want to improve.

4

## Outcome

One capability you want to demonstrate by Day 4.


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# Ground rules

1

## Participate

Ask questions and explain your decisions.

2

## Respect

Give constructive feedback on work, not people.

3

## Protect

Use only authorised, non-confidential inputs.

4

## Verify

Treat generated output as unverified until checked.

5

## Version

Save incremental copies before risky operations.

6

## Evidence

Capture the process, not only the final render.


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FOUR-DAY JOURNEY

# Course architecture

1

## Topic 1

Generate concepts and models with AI support in Blender.

2

## Topic 2

Analyse, validate and refine geometry, UVs and materials.

3

## Topic 3

Optimise, document, export and release production assets.

4

## Assessment

Six SAQs and three integrated practical tasks.


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SCHEDULE

# Lesson plan — Days 1 and 2

## DAY 1

- AI-assisted workflow and model briefs
- Prompt architecture and references
- Blender scene structure and bpy
- Activities 1–3

## DAY 2

- Variation and responsible use
- Validation and topology
- Mesh repair and evidence
- Activities 4–6


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SCHEDULE

# Lesson plan — Days 3 and 4

## DAY 3

- Retopology and polygon budgets
- UVs and PBR materials
- AI-assisted textures and lighting
- Activities 7–8

## DAY 4

- Standards, rendering and export
- Game and print workflows
- Capstone review package
- 3-hour WA + PP


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BY THE END OF THE COURSE

# Learning outcomes

1

## LO1

Devise metrics for AI-assisted 3D design outputs and formulate solutions to modelling problems.

2

## LO2

Analyse Blender models for accuracy, topology quality and alignment with design requirements.

3

## LO3

Review and prepare 3D assets for organisational needs, applicable standards and production delivery.

4

## Evidence

Every outcome is demonstrated through progressive Blender activities.


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# TSC alignment

1

## K1–K3

Conventions, digital methods, drawing/model types and specifications.

2

## K4–K6

Standards, design documentation and enabling IT concepts.

3

## A1–A2

Define quality metrics and solve modelling problems.

4

## A3–A6

Review accuracy, construction needs, strategy and conformance.


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READ BEFORE ASSESSMENT

# Briefing for assessment

1

## Individual

No discussion; complete your own submission.

2

## Open book

Use slides, Learner Guide and approved materials only.

3

## Devices

Place phones and other unapproved materials under the table.

4

## Integrity

No photography or recording of assessment scripts.

5

## Writing

Use black/blue pen where hard copy is used; no correction fluid.

6

## Submission

Upload the completed document to the LMS.


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# Final assessment

## Written Assessment

- 6 open-ended SAQs
- K1–K6 coverage
- 1 hour · open book

## Practical Performance

- 3 integrated Blender tasks
- A1–A6 coverage
- 2 hours · open book

## Competency

- Assessment attendance
- Competent / Not Yet Competent
- Appeal and reassessment available


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ON ASSESSMENT DAY

# Assessment flow

1. TRAQOM / SSG attendance
2. Written Assessment (SAQ)
3. Practical Performance (PP)
4. Upload answers to LMS
5. Sign Assessment Summary Record


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SELF-CONTAINED FOLDERS

# Access the activity packs

## Option A

- Open the GitHub repository
- Clone with Git
- Work from activities/NN-title

## Option B

- GitHub Code menu
- Download ZIP
- Extract before opening files

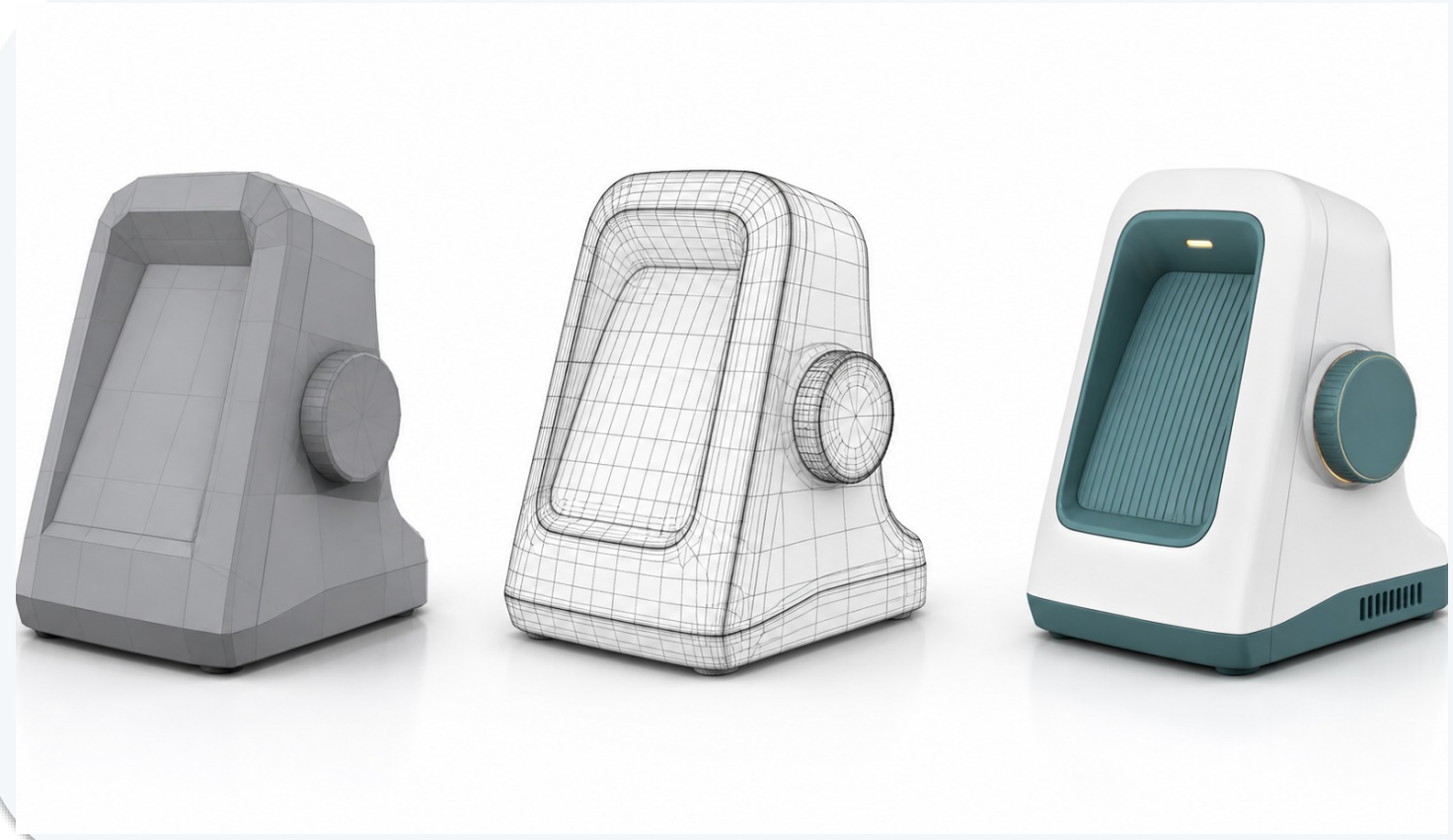
## Evidence

- Follow INSTRUCTIONS.md
- Use CHECKLIST.md
- Update Example-Presentation.pptx

**https://github.com/tertiarycourses/TGS-2023037544-Generative-AI-for-3D-Modeling**


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# From blockout to production-ready asset

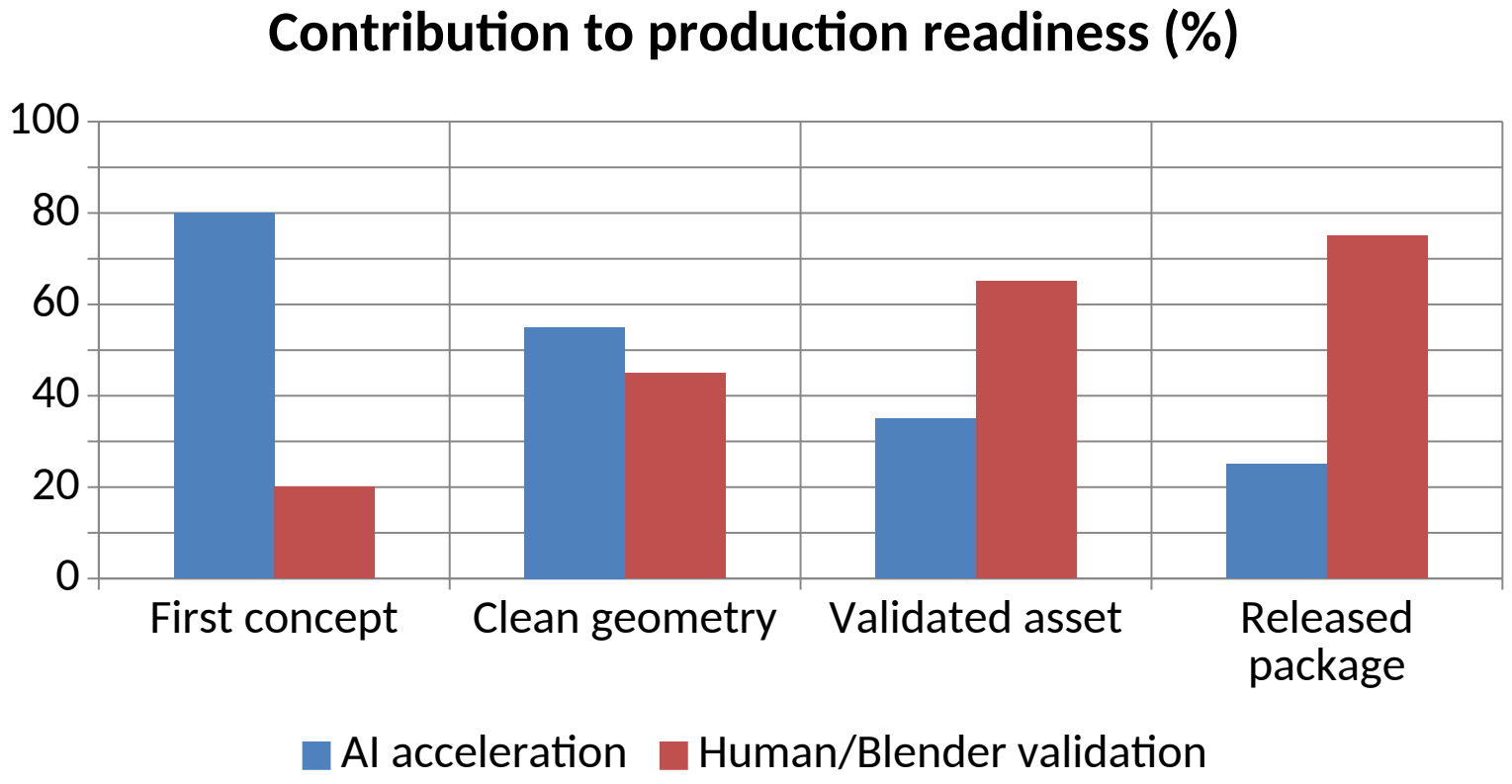


**Compare the same design at three maturity levels: blockout, topology-controlled model and finished material render. Each level answers a different review question.**

Concept visual — validate against the editable Blender file.


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# Where production readiness comes from




WHAT THE DATA SHOWS

**AI assistance can accelerate the first model, but production readiness comes from Blender validation, repair and evidence.**

**The course measures both speed and conformance.**


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# Source-informed build

1

## Legacy deck

Preserved TSC, assessment and technical-drawing coverage as the floor.

2

## Course page

Aligned the four-day course, outcomes and three topic structure.

3

## Video

Adopted the ChatGPT-to-Blender script loop and error-feedback practice.

4

## Blender Manual

Updated terminology and production workflow around Blender.


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TOPIC 1

01

# Generative AI for 3D Design Concepts and Modeling

Briefs · prompts · Blender foundations · script-assisted construction · design variants


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PROGRESSIVE LEARNING

# Topic 1 — capability map

1

## Understand

Briefs

2

## Apply

prompts

3

## Review

K1 · K2 · K3 · K6 · A1 · A2

4

## Demonstrate

Activities 1–4


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# AI-assisted 3D workflow

1

## Brief

Define purpose, audience, dimensions and constraints.

2

## Generate

Produce concepts, scripts or reference images.

3

## Refine

Edit geometry, topology and materials in Blender.

4

## Validate

Measure, inspect and document before release.


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# Decisions and risks — AI-assisted 3D workflow

## DECISIONS THAT IMPROVE QUALITY

- Choose the AI contribution before choosing a tool.
- Keep a human approval gate after every generated output.
- Preserve prompt, version and source evidence.
- Treat generated geometry as unverified input.

## FAILURE MODES TO CONTROL

- Plausible-looking geometry may be structurally unusable.
- Generated scripts can reference outdated API calls.
- Untracked iterations make decisions impossible to audit.


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# AI-assisted 3D workflow

PRINCIPLE

**Generative AI accelerates ideation and repetitive construction; Blender remains the environment for inspection, correction and accountable delivery.**

APPLIED EXAMPLE

A product designer asks AI for a Blender blockout script, runs it in a duplicate file, then checks scale, silhouette and topology before continuing.

EVIDENCE TO TRACK

1. Time to first valid blockout
2. Percentage of validation checks passed
3. Number of repair iterations


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# From requirement to model brief

1

## Purpose

Game prop, print, visualisation, animation or marketing.

2

## Geometry

Dimensions, proportions, silhouette and required moving parts.

3

## Appearance

Material family, colour, finish and lighting context.

4

## Delivery

File format, polygon budget, scale and evidence.


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# Decisions and risks — From requirement to model brief

## DECISIONS THAT IMPROVE QUALITY

- Separate must-have constraints from creative preferences.
- State units and coordinate assumptions explicitly.
- Define acceptance evidence before generation.
- Identify what must remain editable.

## FAILURE MODES TO CONTROL

- Missing units create scale errors.
- Visual adjectives without geometry cues produce inconsistent variants.
- Unclear delivery formats cause rework.


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# From requirement to model brief

PRINCIPLE

**A precise model brief converts ambiguous intent into testable geometry, appearance and delivery requirements.**

APPLIED EXAMPLE

Instead of 'make a modern speaker', specify a 180 mm desktop speaker, rounded enclosure, front grille, separate control knob, neutral materials and glTF delivery.

EVIDENCE TO TRACK

1. Brief completeness score
2. Number of ambiguous requirements
3. Acceptance criteria coverage


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# Prompt architecture for geometry

1

## Object

Name the object and its functional parts.

2

## Form

Specify primitives, proportions and symmetry.

3

## Topology

State manifold, quad flow or low-poly constraints.

4

## Output

Request a Blender Python script or a reference image deliberately.


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# Decisions and risks — Prompt architecture for geometry

## DECISIONS THAT IMPROVE QUALITY

- Use Blender terms when asking for Blender actions.
- Break complex objects into reusable components.
- Request named objects and collections.
- Ask for safe, rerunnable scripts.

## FAILURE MODES TO CONTROL

- One-sentence 'magic wand' prompts hide constraints.
- Overly complex requests increase script failure points.
- Visual style words do not guarantee printable geometry.


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# Prompt architecture for geometry

PRINCIPLE

**Effective prompts describe topology-relevant structure, not only visual style.**

APPLIED EXAMPLE

A useful request names a body, bezel, knob and feet; states dimensions; requests bevel and weighted-normal modifiers; and asks the script to save no files automatically.

EVIDENCE TO TRACK

1. Required parts created
2. Object naming compliance
3. Script rerun success


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# Reference-image literacy

1

## View

Identify perspective, focal length and occluded surfaces.

2

## Scale

Find a known dimension or reference object.

3

## Symmetry

Decide what can be mirrored.

4

## Uncertainty

Mark inferred surfaces and assumptions.


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# Decisions and risks — Reference-image literacy

## DECISIONS THAT IMPROVE QUALITY

- Prefer orthographic front/side/top references when available.
- Do not trace perspective distortion as real geometry.
- Document assumptions for unseen surfaces.
- Use multiple images to triangulate form.

## FAILURE MODES TO CONTROL

- Perspective can distort proportions.
- AI-generated references may contain impossible joins.
- Copyright or provenance may be unclear.


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# Reference-image literacy

PRINCIPLE

**A reference image is evidence of appearance, not proof of hidden geometry or real-world dimensions.**

APPLIED EXAMPLE

A single three-quarter product image supports silhouette and material decisions, but the underside, wall thickness and exact depth remain assumptions.

EVIDENCE TO TRACK

1. Views available
2. Known dimensions
3. Assumptions recorded


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# Blender scene and data model

1

## Object

Transform container positioned in the scene.

2

## Mesh data

Vertices, edges and faces shared by an object.

3

## Collection

Logical grouping for visibility and export.

4

## Scene

World, camera, render and unit settings.


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# Decisions and risks — Blender scene and data model

## DECISIONS THAT IMPROVE QUALITY

- Name objects by purpose, not primitive type.
- Use collections for source, working and approved states.
- Apply transforms only when the downstream operation requires it.
- Save incremental versions.

## FAILURE MODES TO CONTROL

- Renaming only the mesh data can leave confusing object names.
- Unapplied scale can distort bevels and exports.
- Linked data can cause edits to affect multiple objects.


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# Blender scene and data model

PRINCIPLE

**Blender separates objects, object data, materials, collections and scene settings; understanding that structure prevents destructive edits.**

APPLIED EXAMPLE

AI_Device_BODY and AI_Device_KNOB live in a WORKING collection; a duplicate approved collection is locked before risky cleanup.

EVIDENCE TO TRACK

1. Named-object compliance
2. Collection organisation
3. Transform hygiene


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# Coordinate systems, units and scale

1

## Units

Metric or imperial with an explicit scale.

2

## Axes

Blender Z-up; target engines may use Y-up.

3

## Origin

Defines placement, rotation and export behaviour.

4

## Transforms

Location, rotation and scale must be deliberate.


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# Decisions and risks — Coordinate systems, units and scale

## DECISIONS THAT IMPROVE QUALITY

- Set scene units before dimensional work.
- Place origins at meaningful assembly or pivot points.
- Apply scale before bevel, physics and export when appropriate.
- Test import in the target system.

## FAILURE MODES TO CONTROL

- 1000× size errors between metres and millimetres.
- Wrong pivot breaks animation or placement.
- Negative scale can flip normals.


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# Coordinate systems, units and scale

PRINCIPLE

**A model can look correct in the viewport and still fail in manufacturing or another application because units and axes are wrong.**

APPLIED EXAMPLE

A 180 mm prototype is modeled as 0.18 m, origin at its base centre, scale 1/1/1, and export conversion documented.

EVIDENCE TO TRACK

1. Bounding-box dimensions
2. Transform values
3. Round-trip scale error


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# Mesh anatomy and selection

1

## Vertices

Points that define positions.

2

## Edges

Connections controlling flow and boundaries.

3

## Faces

Surface polygons with winding and normals.

4

## Loops

Continuous selections used for form and topology control.


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# Decisions and risks — Mesh anatomy and selection

## DECISIONS THAT IMPROVE QUALITY

- Choose object mode for whole-object transforms.
- Use edit mode for component-level changes.
- Select by topology, not only by screen rectangle.
- Inspect hidden and internal geometry.

## FAILURE MODES TO CONTROL

- Accidental duplicate vertices create shading and manifold defects.
- Hidden internal faces increase polygon count.
- Random edge flow causes pinching.


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# Mesh anatomy and selection

PRINCIPLE

**Vertices, edges, faces, loops and islands are the editable structure that turns an AI concept into controllable geometry.**

APPLIED EXAMPLE

A supporting loop around a product opening controls its edge sharpness under subdivision without adding an unrelated separate object.

EVIDENCE TO TRACK

1. Non-manifold elements
2. Duplicate vertices
3. Edge-flow continuity


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# Primitive-based blockout

1

## Box

Hard-surface bodies and enclosures.

2

## Cylinder

Knobs, ports and radial features.

3

## Plane

Profiles, panels and reference surfaces.

4

## Curve

Cables, trims and controlled sweeps.


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# Decisions and risks — Primitive-based blockout

## DECISIONS THAT IMPROVE QUALITY

- Validate silhouette from multiple views.
- Keep components separate while proportions change.
- Use symmetry before manual duplication.
- Delay bevel and micro-detail.

## FAILURE MODES TO CONTROL

- Detail can hide incorrect proportion.
- Boolean-heavy blockouts become brittle if unmanaged.
- Joined meshes are harder to revise.


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# Primitive-based blockout

PRINCIPLE

**Simple primitives establish proportion and hierarchy faster and more safely than prematurely detailed meshes.**

APPLIED EXAMPLE

A desktop device begins as a beveled box, a cylinder knob and a recessed panel; the team approves silhouette before adding vents.

EVIDENCE TO TRACK

1. Silhouette approval
2. Primary dimension error
3. Number of editable components


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# Non-destructive modifiers

1

## Mirror

Symmetric construction from one side.

2

## Bevel

Manufacturable edge treatment and highlights.

3

## Subdivision

Smooth form from a control cage.

4

## Solidify

Thickness from a surface.


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# Decisions and risks — Non-destructive modifiers

## DECISIONS THAT IMPROVE QUALITY

- Order modifiers by dependency.
- Use conservative viewport levels for performance.
- Name critical modifiers.
- Apply only at a controlled handoff point.

## FAILURE MODES TO CONTROL

- Wrong stack order changes geometry.
- Applying too early destroys editability.
- Unapplied scale changes bevel width.


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# Non-destructive modifiers

PRINCIPLE

**A modifier stack records intent and preserves editability while transforming the visible result.**

APPLIED EXAMPLE

Mirror → Bevel → Weighted Normal produces a symmetric hard-surface body with controlled highlights and editable dimensions.

EVIDENCE TO TRACK

1. Modifier count
2. Stack order check
3. Applied-vs-live status


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# Blender Python and bpy

1

## Data API

Direct access to objects, meshes, materials and scenes.

2

## Operators

Context-sensitive actions similar to UI commands.

3

## Idempotence

A safe rerun reaches the same intended state.

4

## Logging

Names and messages make failures diagnosable.


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# Decisions and risks — Blender Python and bpy

## DECISIONS THAT IMPROVE QUALITY

- Ask the model to target the installed Blender version.
- Delete or reuse only named generated objects.
- Avoid automatic overwrite or save calls.
- Run in a copy and inspect the console.

## FAILURE MODES TO CONTROL

- Context-sensitive operators fail when the wrong mode is active.
- Generated code may call deprecated properties.
- Broad delete commands can destroy learner work.


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# Blender Python and bpy

PRINCIPLE

**Blender's Python API can create and edit data deterministically, but generated scripts must be scoped, readable and reversible.**

APPLIED EXAMPLE

A script removes only objects in collection AI_GENERATED, rebuilds the blockout, names every part and prints a completion summary.

EVIDENCE TO TRACK

1. Script run success
2. Objects created as specified
3. Destructive operations reviewed


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# Error-feedback iteration

1

## Capture

Copy the full traceback and Blender version.

2

## Classify

Syntax, missing data, context or API mismatch.

3

## Repair

Change the smallest relevant section.

4

## Retest

Run in a clean duplicate and compare results.


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# Decisions and risks — Error-feedback iteration

## DECISIONS THAT IMPROVE QUALITY

- Do not ask AI to rewrite everything after one error.
- Provide the exact error and intended result.
- Inspect the proposed fix before running.
- Record successful changes.

## FAILURE MODES TO CONTROL

- Fixes can introduce silent geometry changes.
- Partial runs may leave duplicate objects.
- Copying only the last error line loses context.


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# Error-feedback iteration

PRINCIPLE

**An error message is structured evidence: capture the traceback, locate the failing line, form a hypothesis and test one repair at a time.**

APPLIED EXAMPLE

A context error from bpy.ops is repaired by switching to direct data creation rather than repeatedly changing selection state.

EVIDENCE TO TRACK

1. Repair iterations
2. Regression checks passed
3. Duplicate object count


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# Design variation and selection

1

## Diverge

Generate meaningful alternatives.

2

## Normalize

Use the same view, scale and lighting.

3

## Score

Apply weighted criteria.

4

## Converge

Select and record the decision.


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# Decisions and risks — Design variation and selection

## DECISIONS THAT IMPROVE QUALITY

- Vary one major design assumption per variant.
- Use common camera and materials for fair comparison.
- Weight functional criteria above novelty.
- Preserve rejected variants with reasons.

## FAILURE MODES TO CONTROL

- Beauty bias overrides requirements.
- Unequal renders make comparisons unfair.
- Decision rationale is lost.


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# Design variation and selection

PRINCIPLE

**Generative breadth is valuable only when variants are compared against explicit requirements.**

APPLIED EXAMPLE

Three enclosures are scored on usability, manufacturability, brand fit and production cost; the winner is not simply the most visually novel.

EVIDENCE TO TRACK

1. Criteria weighted
2. Variant score spread
3. Decision record completeness


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# Prompt and asset provenance

1

## Source

Reference owner, licence and retrieval date.

2

## Prompt

Exact request and model/tool context.

3

## Changes

Manual fixes and derived versions.

4

## Approval

Reviewer, date and accepted purpose.


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# Decisions and risks — Prompt and asset provenance

## DECISIONS THAT IMPROVE QUALITY

- Store prompts beside the activity evidence.
- Mark generated reference images as concept visuals.
- Separate reusable source from final derivative work.
- Never assume AI output is rights-cleared.

## FAILURE MODES TO CONTROL

- Unknown training/source provenance.
- Unlicensed reference imagery.
- Missing modification history.


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# Prompt and asset provenance

PRINCIPLE

**A production asset needs traceability across input references, prompts, generated outputs, manual edits and approvals.**

APPLIED EXAMPLE

The asset register links the prompt, concept image, Blender file v03, validation checklist and final export.

EVIDENCE TO TRACK

1. Assets with complete provenance
2. Approvals recorded
3. Unresolved rights questions


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# Human review and responsible use

1

## Accuracy

Does the model match dimensions and function?

2

## Safety

Could geometry create hazards or misleading claims?

3

## Rights

Are references and outputs legally usable?

4

## Disclosure

Would a stakeholder mistake concept imagery for verified evidence?


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# Decisions and risks — Human review and responsible use

## DECISIONS THAT IMPROVE QUALITY

- Label concept visuals clearly.
- Escalate safety-critical and regulated uses.
- Remove sensitive information from prompts.
- Require human sign-off before release.

## FAILURE MODES TO CONTROL

- Hallucinated mechanisms.
- Confidential designs exposed to external services.
- Synthetic renders misrepresented as real prototypes.


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# Human review and responsible use

PRINCIPLE

**AI output must be reviewed for safety, bias, plausibility, intellectual property and fitness for the stated use.**

APPLIED EXAMPLE

A generated medical-device concept may support ideation but cannot be represented as engineered, tested or clinically approved.

EVIDENCE TO TRACK

1. Review gates completed
2. Sensitive fields removed
3. Disclosure present


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# Evidence-driven activity practice

1

## Brief

What was requested.

2

## Process

Prompt, settings and revisions.

3

## Evidence

Screenshots and measurements.

4

## Acceptance

Checklist and reflection.


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# Decisions and risks — Evidence-driven activity practice

## DECISIONS THAT IMPROVE QUALITY

- Capture before/after from the same view.
- Include Blender file and exported output.
- Name evidence consistently.
- Explain one issue and its fix.

## FAILURE MODES TO CONTROL

- Screenshots without context.
- Final image with no editable file.
- No evidence that acceptance tests were run.


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# Evidence-driven activity practice

PRINCIPLE

**A strong activity submission proves process, decision quality and acceptance—not just a final attractive screenshot.**

APPLIED EXAMPLE

The sample presentation shows the brief, AI contribution, Blender validation, final render and a signed acceptance checklist.

EVIDENCE TO TRACK

1. Required evidence items
2. Acceptance checks passed
3. Reflection quality


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HANDS-ON PRACTICE

# Topic 1 activities

## Activities 1–2

- Translate a Design Requirement into an AI-Ready 3D Brief
- Generate a Blender Gold-Sphere Scene with AI-Assisted Python

## Activities 3–4

- Iterate and Repair an AI-Generated Blender Script
- Develop and Select Three AI-Assisted Product Variants

## Progression

- Brief or inspect
- Build or repair
- Verify and present


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# Translate a Design Requirement into an AI-Ready 3D Brief

ACTIVITY 1

**A retail innovation team needs a compact desktop product visual for stakeholder review and later prototyping.**

DELIVERABLE

**A one-page model brief, prompt specification and acceptance scorecard.**

ALIGNMENT

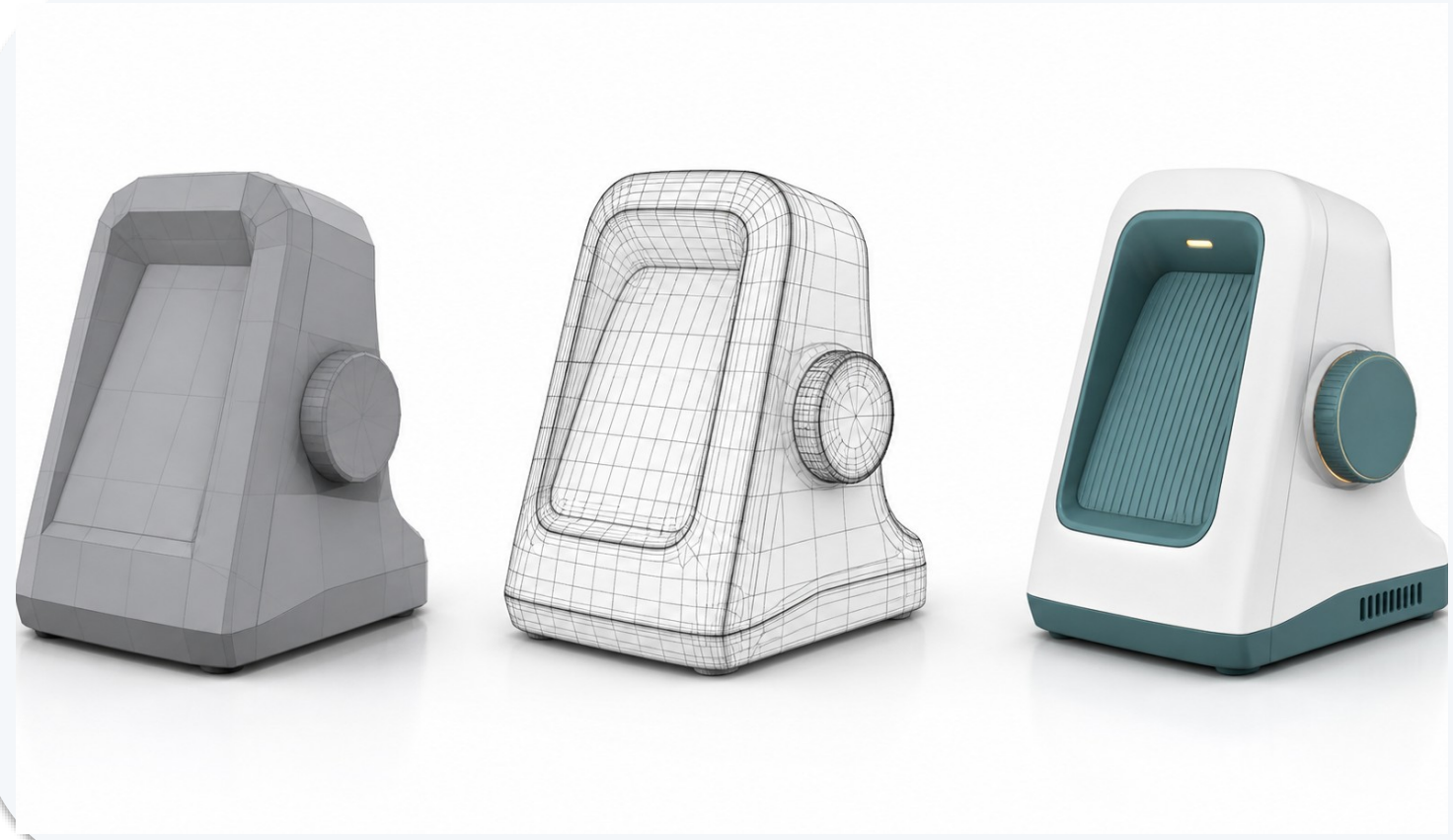
**K1 · K3 · A1**

OBJECTIVE

Create measurable geometry, appearance and delivery criteria before using AI.


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# Activity 1 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 1 — quality gates

## Form

- Purpose and audience are explicit
- Dimensions and units are stated

## Structure

- Editable parts are named
- Acceptance criteria are measurable

## Delivery

- Delivery formats are defined
- A one-page model brief, prompt specification and acceptance scorecard.


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# Activity 1 — acceptance evidence

## ACCEPTANCE CHECKS

- Purpose and audience are explicit
- Dimensions and units are stated
- Editable parts are named
- Acceptance criteria are measurable
- Delivery formats are defined

## EVIDENCE TO SUBMIT

- Completed model brief
- Prompt specification
- Acceptance scorecard
- One-slide decision summary


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# Generate a Blender Gold-Sphere Scene with AI-Assisted Python

ACTIVITY 2

**The team wants a repeatable proof-of-concept showing how natural-language intent can create named geometry and a material in Blender.**

DELIVERABLE

**A safe Blender script, a generated gold sphere scene and an execution evidence presentation.**

ALIGNMENT

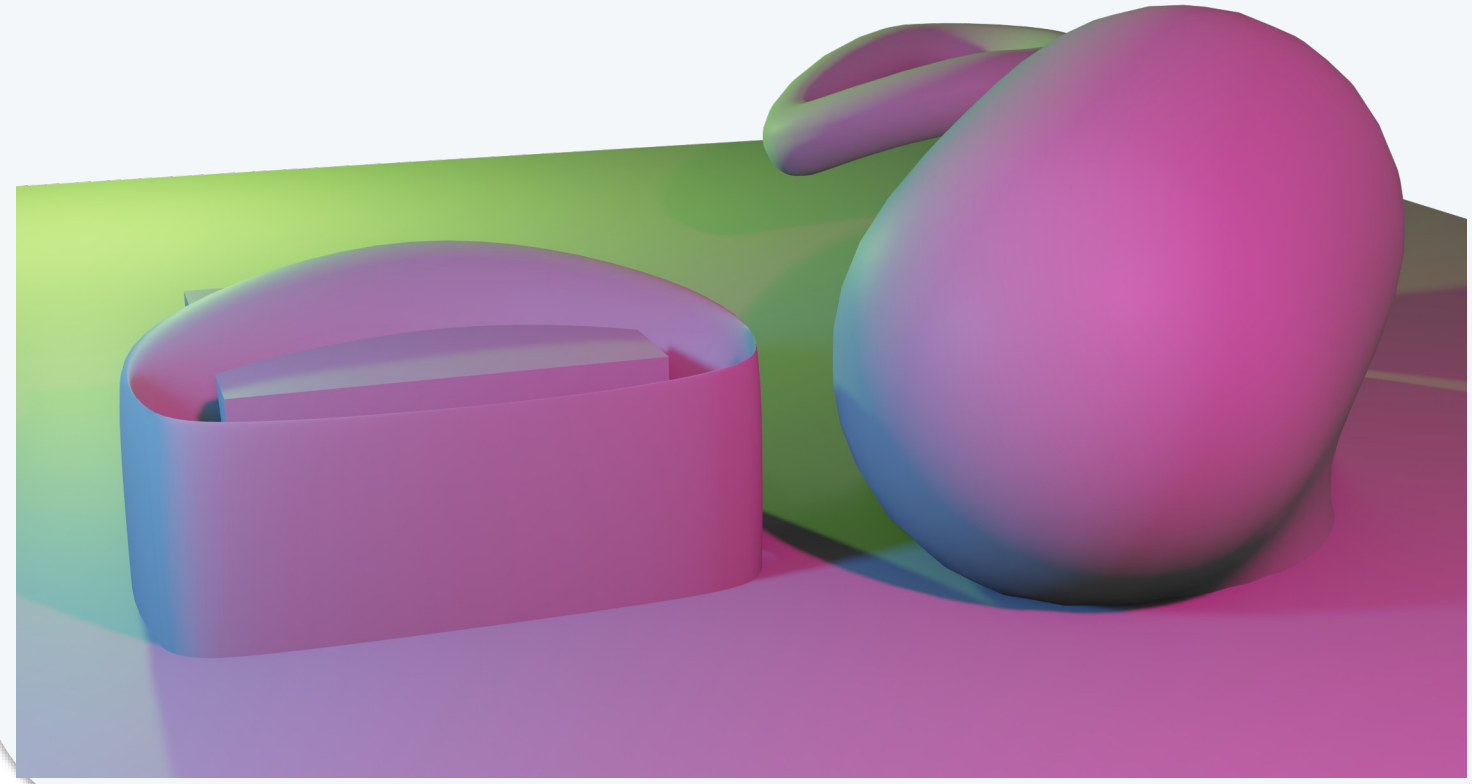
**K2 · K6 · A2**

OBJECTIVE

Use the video workflow to request, inspect, run and verify a small Blender Python script.


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# Activity 2 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 2 — quality gates

## Form

- Script targets the installed Blender version
- No automatic file overwrite

## Structure

- Named sphere and material exist
- Scale and placement verified

## Delivery

- Render and Blender source saved
- A safe Blender script, a generated gold sphere scene and an execution evidence presentation.


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# Activity 2 — acceptance evidence

## ACCEPTANCE CHECKS

- Script targets the installed Blender version
- No automatic file overwrite
- Named sphere and material exist
- Scale and placement verified
- Render and Blender source saved

## EVIDENCE TO SUBMIT

- Exact prompt
- Reviewed script
- Outliner screenshot
- Rendered result
- Completed .blend file


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# Iterate and Repair an AI-Generated Blender Script

ACTIVITY 3

**A generated script fails after a Blender update and leaves partial objects in the scene.**

DELIVERABLE

**A repaired rerunnable script, an error log and a before/after diagnosis.**

ALIGNMENT

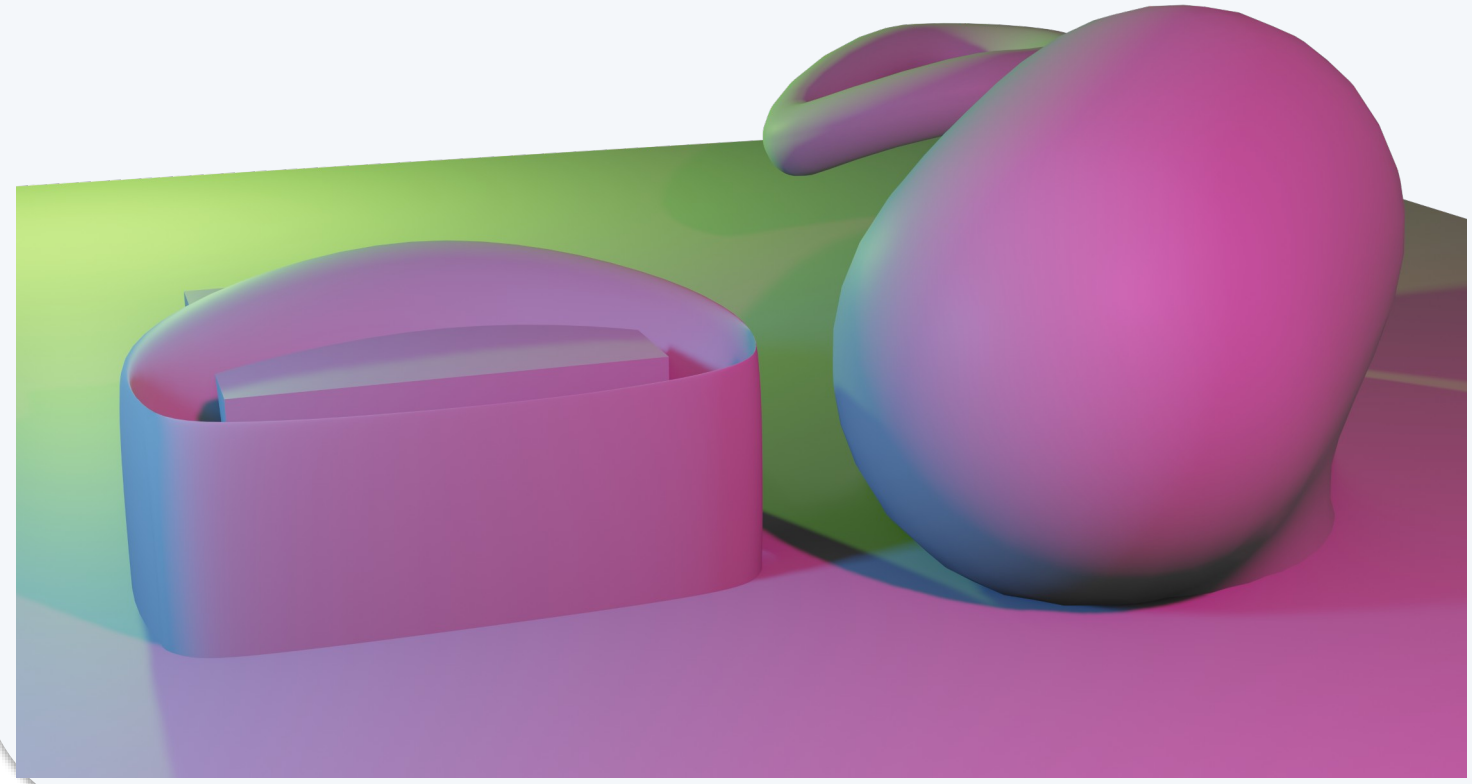
**K2 · K6 · A2**

OBJECTIVE

Diagnose a controlled script failure and use evidence-driven AI feedback to repair it safely.


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# Activity 3 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 3 — quality gates

## Form

- Full traceback captured
- Root cause classified

## Structure

- Patch is narrowly scoped
- Second run creates no duplicates

## Delivery

- Expected geometry and modifiers verified
- A repaired rerunnable script, an error log and a before/after diagnosis.


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# Activity 3 — acceptance evidence

## ACCEPTANCE CHECKS

- Full traceback captured
- Root cause classified
- Patch is narrowly scoped
- Second run creates no duplicates
- Expected geometry and modifiers verified

## EVIDENCE TO SUBMIT

- Error screenshot
- Root-cause statement
- Before/after code excerpt
- Two-run Outliner check
- Final script and .blend


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# Develop and Select Three AI-Assisted Product Variants

ACTIVITY 4

**Stakeholders want options for the desktop product but need a defendable recommendation, not a gallery of arbitrary shapes.**

DELIVERABLE

**Three normalized Blender blockouts, a weighted decision matrix and a selected direction.**

ALIGNMENT

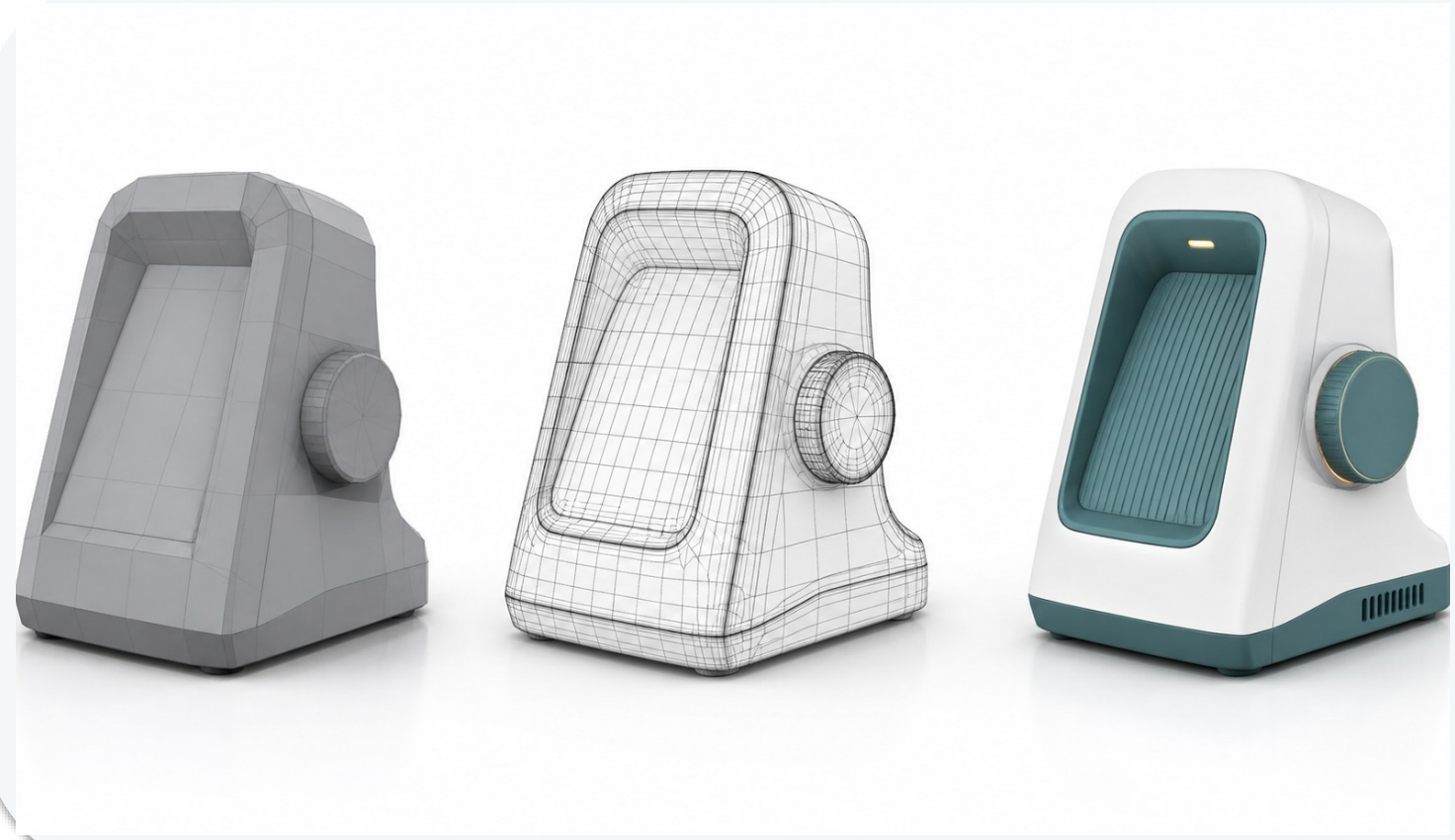
**K1 · K3 · A1 · A5**

OBJECTIVE

Generate meaningful design alternatives and select one using weighted acceptance criteria.


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# Activity 4 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 4 — quality gates

## Form

- Three variants differ meaningfully
- Common camera and lighting used

## Structure

- Criteria total 100%
- Scores cite evidence

## Delivery

- Recommendation includes trade-off
- Three normalized Blender blockouts, a weighted decision matrix and a selected direction.


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# Activity 4 — acceptance evidence

## ACCEPTANCE CHECKS

- Three variants differ meaningfully
- Common camera and lighting used
- Criteria total 100%
- Scores cite evidence
- Recommendation includes trade-off

## EVIDENCE TO SUBMIT

- Nine comparable views
- Weighted matrix
- Sensitivity check
- Recommendation and mitigation
- Variant collections


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WHAT YOU CAN NOW DEMONSTRATE

# Topic 1 recap

1

## Concepts

15 decision-ready frameworks

2

## Activities

Activities 1–4 completed

3

## Evidence

Editable Blender files, metrics and presentations

4

## Alignment

K1 · K2 · K3 · K6 · A1 · A2


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TOPIC 2

02

# AI-Assisted 3D Model Analysis, Validation and Refinement

Inspection · topology · normals · retopology · UVs · materials · controlled repair


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PROGRESSIVE LEARNING

# Topic 2 — capability map

1

## Understand

Inspection

2

## Apply

topology

3

## Review

K1 · K4 · K5 · A2 · A3 · A4

4

## Demonstrate

Activities 5–8


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# Validation framework

1

## Form

Silhouette and proportions.

2

## Structure

Topology, normals and manifold status.

3

## Appearance

UVs, materials and shading.

4

## Delivery

Naming, scale, format and evidence.


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# Decisions and risks — Validation framework

## DECISIONS THAT IMPROVE QUALITY

- Validate from broad shape to small defects.
- Measure against the brief, not memory.
- Separate blocking defects from refinements.
- Record pass/fail evidence.

## FAILURE MODES TO CONTROL

- A render can conceal bad topology.
- One viewport angle can hide proportion errors.
- Unwritten checks are applied inconsistently.


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# Validation framework

PRINCIPLE

**Validation checks form, structure, dimensions, appearance and delivery requirements in a repeatable order.**

APPLIED EXAMPLE

A visually appealing model fails because its wall is open and its scale is 1000× too large; validation catches both before export.

EVIDENCE TO TRACK

1. Checks passed
2. Blocking defects
3. Review time


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# Silhouette and proportion review

1

## Front

Width, symmetry and feature alignment.

2

## Side

Depth, taper and profile.

3

## Top

Footprint and spacing.

4

## Perspective

Overall visual coherence.


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# Decisions and risks — Silhouette and proportion review

## DECISIONS THAT IMPROVE QUALITY

- Compare against calibrated references.
- Use overlays or guides for key dimensions.
- Check negative space as carefully as solids.
- Review at target viewing distance.

## FAILURE MODES TO CONTROL

- Perspective bias.
- AI reference inconsistency across views.
- Feature creep changes primary proportions.


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# Silhouette and proportion review

PRINCIPLE

**The silhouette communicates primary form; orthographic views expose proportion errors that perspective views can hide.**

APPLIED EXAMPLE

A handle reads well in perspective but blocks the opening in side view; the review prioritises functional clearance.

EVIDENCE TO TRACK

1. Dimension variance
2. Silhouette agreement
3. Clearance minimum


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# Topology quality

1

## Edge flow

Loops follow form and deformation.

2

## Density

Resolution is allocated where curvature or silhouette needs it.

3

## Poles

Higher-valence vertices are placed away from critical areas.

4

## Faces

Quads, triangles and n-gons are used intentionally.


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# Decisions and risks — Topology quality

## DECISIONS THAT IMPROVE QUALITY

- Topology is purpose-dependent.
- Static hard-surface assets tolerate different flow from characters.
- Inspect subdivision behaviour.
- Avoid hidden internal geometry.

## FAILURE MODES TO CONTROL

- Random triangles cause shading noise.
- Uniform high density wastes performance.
- N-gons can break under deformation or triangulation.


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# Topology quality

PRINCIPLE

**Good topology supports the intended deformation, shading, editing and export while using no more geometry than necessary.**

APPLIED EXAMPLE

A product body uses clean supporting loops at openings and low density on flat panels; a decorative triangulated insert remains separate.

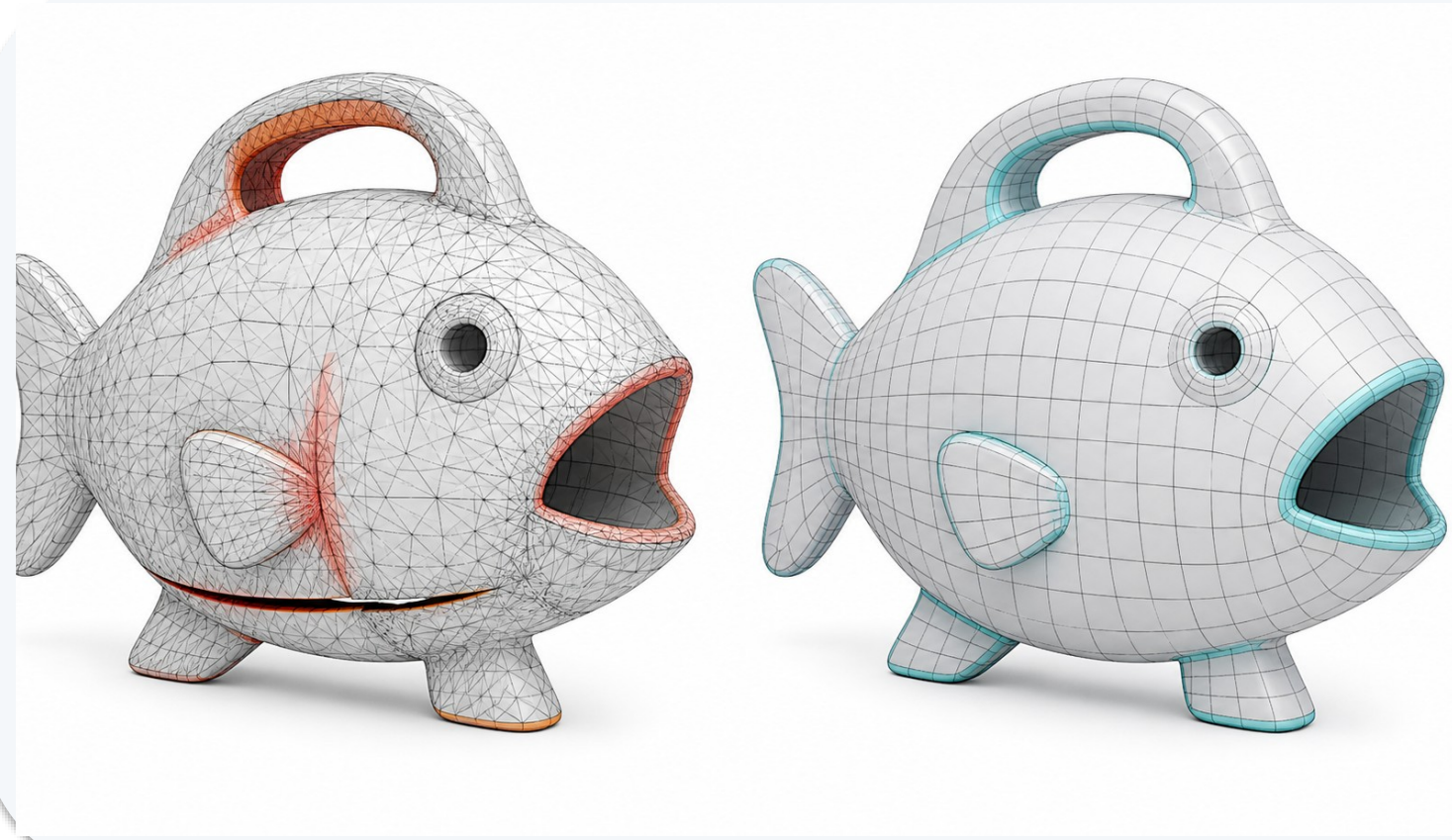
EVIDENCE TO TRACK

1. Polygon count
2. Long-thin face ratio
3. Subdivision artefacts


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CONCEPT VISUAL

# Mesh quality — before and after



**The same conceptual object can look similar in silhouette while differing radically in topology. Validate manifold status, edge flow and shading before approval.**

Concept visual — validate against the editable Blender file.


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# Manifold and watertight geometry

1

## Boundary

Open edges indicate holes.

2

## Non-manifold

Edges shared by the wrong number of faces.

3

## Internal

Faces trapped inside a volume.

4

## Thickness

Walls must meet the target process minimum.


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# Decisions and risks — Manifold and watertight geometry

## DECISIONS THAT IMPROVE QUALITY

- Use Select Non-Manifold as a diagnostic, not an automatic cure.
- Repair small gaps before remeshing.
- Check disconnected shells.
- Recalculate volume after repair.

## FAILURE MODES TO CONTROL

- Automatic fill may create poor n-gons.
- Overlapping shells can appear watertight.
- Zero-thickness surfaces fail fabrication.


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# Manifold and watertight geometry

PRINCIPLE

**A manifold mesh forms an unambiguous closed surface; this is critical for printing, booleans and reliable downstream processing.**

APPLIED EXAMPLE

A fish-shaped container has an unintended seam along the belly; merging boundary vertices and rebuilding faces closes the volume.

EVIDENCE TO TRACK

1. Boundary edge count
2. Connected components
3. Minimum wall thickness


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# Normals and face orientation

1

## Face normals

Outward direction of polygon faces.

2

## Vertex normals

Interpolated direction used for smooth shading.

3

## Auto Smooth

Controls splitting at sharp angles.

4

## Weighted normals

Biases shading toward larger faces.


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# Decisions and risks — Normals and face orientation

## DECISIONS THAT IMPROVE QUALITY

- Enable Face Orientation to spot flipped faces.
- Recalculate outside only after topology is valid.
- Mark sharp edges deliberately.
- Test in the target renderer.

## FAILURE MODES TO CONTROL

- Negative scale flips orientation.
- Duplicate faces create z-fighting.
- Custom normals can hide structural problems.


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# Normals and face orientation

PRINCIPLE

**Normals define the direction surfaces face and influence shading, culling and export.**

APPLIED EXAMPLE

Red face-orientation patches reveal inverted faces near a boolean cut; correction removes black shading after export.

EVIDENCE TO TRACK

1. Flipped faces
2. Shading artefacts
3. Sharp-edge consistency


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# Transform hygiene

1

## Location

Placement relative to origin.

2

## Rotation

Orientation relative to axes.

3

## Scale

Object multiplier over mesh coordinates.

4

## Origin

Pivot and placement reference.


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# Decisions and risks — Transform hygiene

## DECISIONS THAT IMPROVE QUALITY

- Check dimensions before applying scale.
- Keep origin intentional.
- Avoid negative scale for final delivery.
- Document exceptions.

## FAILURE MODES TO CONTROL

- Applying rotation can affect rig or export assumptions.
- Origins reset accidentally.
- Mirrored negative scales cause culling problems.


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# Transform hygiene

PRINCIPLE

**Applied transforms make geometric operations predictable, but application must respect pivots, animation and linked data.**

APPLIED EXAMPLE

A bevel width varies across two objects because one has unapplied non-uniform scale; applying scale restores consistent treatment.

EVIDENCE TO TRACK

1. Scale equals 1/1/1
2. Rotation convention
3. Origin location


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# Cleanup strategy

1

## Diagnose

Identify cause and affected region.

2

## Protect

Duplicate or version the source.

3

## Repair

Merge, dissolve, rebuild or remesh deliberately.

4

## Verify

Repeat the original failing test.


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# Decisions and risks — Cleanup strategy

## DECISIONS THAT IMPROVE QUALITY

- Start with duplicate vertices and internal faces.
- Preserve silhouette while simplifying.
- Use local repair before global remesh.
- Compare statistics before and after.

## FAILURE MODES TO CONTROL

- Aggressive cleanup changes silhouette.
- Dissolve can remove supporting edges.
- Global operations erase UVs and material assignments.


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# Cleanup strategy

PRINCIPLE

**Repair the smallest root cause that restores requirements; do not destroy useful structure with indiscriminate remeshing.**

APPLIED EXAMPLE

A dense AI mesh needs only a local opening repair and decimation of hidden detail—not a complete voxel remesh.

EVIDENCE TO TRACK

1. Defects removed
2. Silhouette deviation
3. Data preserved


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# Retopology approaches

1

## Manual

Highest control for key loops and deformation.

2

## Shrinkwrap

Projects a clean cage onto a source.

3

## Remesh

Fast uniform topology with less semantic control.

4

## Decimate

Reduces faces while retaining triangulated shape.


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# Decisions and risks — Retopology approaches

## DECISIONS THAT IMPROVE QUALITY

- Define target use before selecting a method.
- Protect high-detail source.
- Place loops around deformation and openings.
- Bake detail when reducing density.

## FAILURE MODES TO CONTROL

- Retopology can erase design features.
- Shrinkwrap can project through thin surfaces.
- Automatic remesh may lose hard edges.


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# Retopology approaches

PRINCIPLE

**Retopology rebuilds surface structure for the target use; the right method balances speed, control and preservation.**

APPLIED EXAMPLE

A static visualisation asset may use controlled decimation; an animated creature needs manual quad flow around joints.

EVIDENCE TO TRACK

1. Target polygon budget
2. Feature retention
3. Deformation test


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# Polygon budgets and level of detail

1

## Silhouette

Spend polygons where the outline changes.

2

## Deformation

Support joints and bending.

3

## Materials

Avoid geometry for detail that textures can carry.

4

## LOD

Prepare lower-detail variants for distance.


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# Decisions and risks — Polygon budgets and level of detail

## DECISIONS THAT IMPROVE QUALITY

- Measure triangles after triangulation.
- Benchmark in the target environment.
- Use consistent LOD naming.
- Avoid invisible internal detail.

## FAILURE MODES TO CONTROL

- Viewport performance is not target performance.
- Decimation can break UVs.
- LOD transitions can pop visibly.


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# Polygon budgets and level of detail

PRINCIPLE

**A polygon budget is a performance and delivery constraint tied to screen size, platform and use—not a universal number.**

APPLIED EXAMPLE

A hero product render keeps fine geometry; a real-time background prop uses baked normals and a lower triangle count.

EVIDENCE TO TRACK

1. Triangles per LOD
2. Frame-time contribution
3. Silhouette error


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# UV coordinates and seams

1

## Seams

Cuts placed along hidden or logical boundaries.

2

## Islands

Contiguous UV regions.

3

## Distortion

Stretching between model and map.

4

## Packing

Efficient use of texture space.


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# Decisions and risks — UV coordinates and seams

## DECISIONS THAT IMPROVE QUALITY

- Apply scale before unwrapping.
- Place seams where discontinuity is acceptable.
- Equalise texel density for comparable surfaces.
- Leave padding for mipmaps and baking.

## FAILURE MODES TO CONTROL

- Overlapping islands cause unintended texture reuse.
- Insufficient padding creates bleeding.
- Unapplied scale distorts unwraps.


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# UV coordinates and seams

PRINCIPLE

**UVs map a 3D surface to 2D texture space; seams determine where the surface is cut and unfolded.**

APPLIED EXAMPLE

A product body hides its main seam on the underside and allocates more UV area to the front control panel.

EVIDENCE TO TRACK

1. Average island stretch
2. Texel density
3. UV occupancy


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# PBR materials

1

## Base colour

Diffuse/reflective colour without lighting baked in.

2

## Roughness

Micro-surface spread of reflections.

3

## Metallic

Dielectric versus conductor behaviour.

4

## Normal

Fine directional detail without new silhouette geometry.


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# Decisions and risks — PBR materials

## DECISIONS THAT IMPROVE QUALITY

- Use plausible parameter ranges.
- Do not paint shadows into base colour.
- Keep colour data and non-colour data settings correct.
- Test under neutral lighting.

## FAILURE MODES TO CONTROL

- Incorrect colour space changes values.
- Overstrong normal maps look embossed.
- AI textures can contain lighting and perspective artefacts.


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# PBR materials

PRINCIPLE

**Physically based materials describe how light interacts with a surface through a small set of meaningful parameters.**

APPLIED EXAMPLE

A powder-coated enclosure uses metallic 0, medium-high roughness and subtle normal detail; a metal knob uses metallic 1 with lower roughness.

EVIDENCE TO TRACK

1. Material channel completeness
2. Texture resolution
3. Neutral-light consistency


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# AI-generated textures

1

## Prompt

Describe material, scale and surface condition.

2

## Tile

Remove seams and edge discontinuity.

3

## Derive

Create roughness/normal carefully.

4

## Validate

Inspect repeats, scale and physical plausibility.


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# Decisions and risks — AI-generated textures

## DECISIONS THAT IMPROVE QUALITY

- Request flat, evenly lit material references.
- Remove recognisable logos or text.
- Use separate masks for controlled variation.
- Preserve provenance.

## FAILURE MODES TO CONTROL

- Baked highlights move with the texture.
- Fake normals invert under lighting.
- Hidden copyrighted marks appear in generated detail.


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# AI-generated textures

PRINCIPLE

**AI can accelerate texture ideation, but images must be made tileable, channel-correct, rights-aware and free of baked perspective.**

APPLIED EXAMPLE

A generated brushed-plastic reference is cleaned into a seamless base colour, then roughness is authored from measured visual cues.

EVIDENCE TO TRACK

1. Seam visibility
2. Repeat frequency
3. Channel plausibility


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# Lighting as a validation tool

1

## Key

Defines form and main highlight.

2

## Fill

Controls contrast and shadow readability.

3

## Rim

Separates silhouette from background.

4

## Environment

Provides consistent reflections.


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# Decisions and risks — Lighting as a validation tool

## DECISIONS THAT IMPROVE QUALITY

- Validate under a neutral studio setup.
- Rotate light to reveal shading artefacts.
- Use consistent exposure across variants.
- Separate evidence renders from marketing renders.

## FAILURE MODES TO CONTROL

- Dramatic shadows conceal geometry defects.
- Different exposure biases variant comparison.
- HDRI licence or provenance is unknown.


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# Lighting as a validation tool

PRINCIPLE

**Neutral lighting reveals material and surface quality; dramatic lighting should come after defects are understood.**

APPLIED EXAMPLE

A grazing key light exposes pinching around a boolean that was invisible under flat viewport shading.

EVIDENCE TO TRACK

1. Exposure consistency
2. Artefacts discovered
3. Variant lighting parity


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# Change control and review

1

## Baseline

Freeze the incoming AI model.

2

## Change

Name the repair and reason.

3

## Compare

Use consistent views and metrics.

4

## Approve

Record reviewer and acceptance.


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# Decisions and risks — Change control and review

## DECISIONS THAT IMPROVE QUALITY

- Use incremental filenames.
- Avoid overwriting the raw source.
- Keep a change log.
- Recheck downstream effects.

## FAILURE MODES TO CONTROL

- Silent overwrite removes evidence.
- A topology fix can break UVs.
- Late changes bypass acceptance tests.


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# Change control and review

PRINCIPLE

**Every major repair should be reviewable against the previous version and the design brief.**

APPLIED EXAMPLE

v01 raw → v02 manifold repair → v03 retopology → v04 UV/material; each version records changed statistics.

EVIDENCE TO TRACK

1. Versions traceable
2. Regression checks
3. Approval status


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# Quality dashboard

1

## Geometry

Dimensions, manifold, normals and polygon count.

2

## Appearance

UV stretch, materials and render consistency.

3

## Delivery

Naming, format and round-trip test.

4

## Risk

Open issues, owner and disposition.


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# Decisions and risks — Quality dashboard

## DECISIONS THAT IMPROVE QUALITY

- Use thresholds tied to the brief.
- Show current value and status.
- Link evidence to each result.
- Never hide a failed check in an average score.

## FAILURE MODES TO CONTROL

- Vanity scores obscure blockers.
- Thresholds without rationale are arbitrary.
- Evidence links point to stale versions.


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# Quality dashboard

PRINCIPLE

**A concise dashboard turns technical checks into an approval decision stakeholders can understand.**

APPLIED EXAMPLE

The dashboard marks 11/12 checks passed and leaves 'minimum wall thickness' amber pending a print-process decision.

EVIDENCE TO TRACK

1. Pass rate
2. Blocking issues
3. Evidence freshness


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# Topic 2 activities

## Activities 5–6

- Inspect an AI-Generated Mesh and Build a Validation Baseline
- Repair Non-Manifold Geometry, Normals and Duplicate Data

## Activities 7–8

- Retopologize and Optimize an AI Mesh for Its Target Use
- UV-Unwrap and Apply AI-Assisted PBR Materials

## Progression

- Brief or inspect
- Build or repair
- Verify and present


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# Inspect an AI-Generated Mesh and Build a Validation Baseline

ACTIVITY 5

**An external AI mesh arrives with no quality record and must be assessed before refinement.**

DELIVERABLE

**A frozen baseline Blender file, defect register and validation dashboard.**

ALIGNMENT

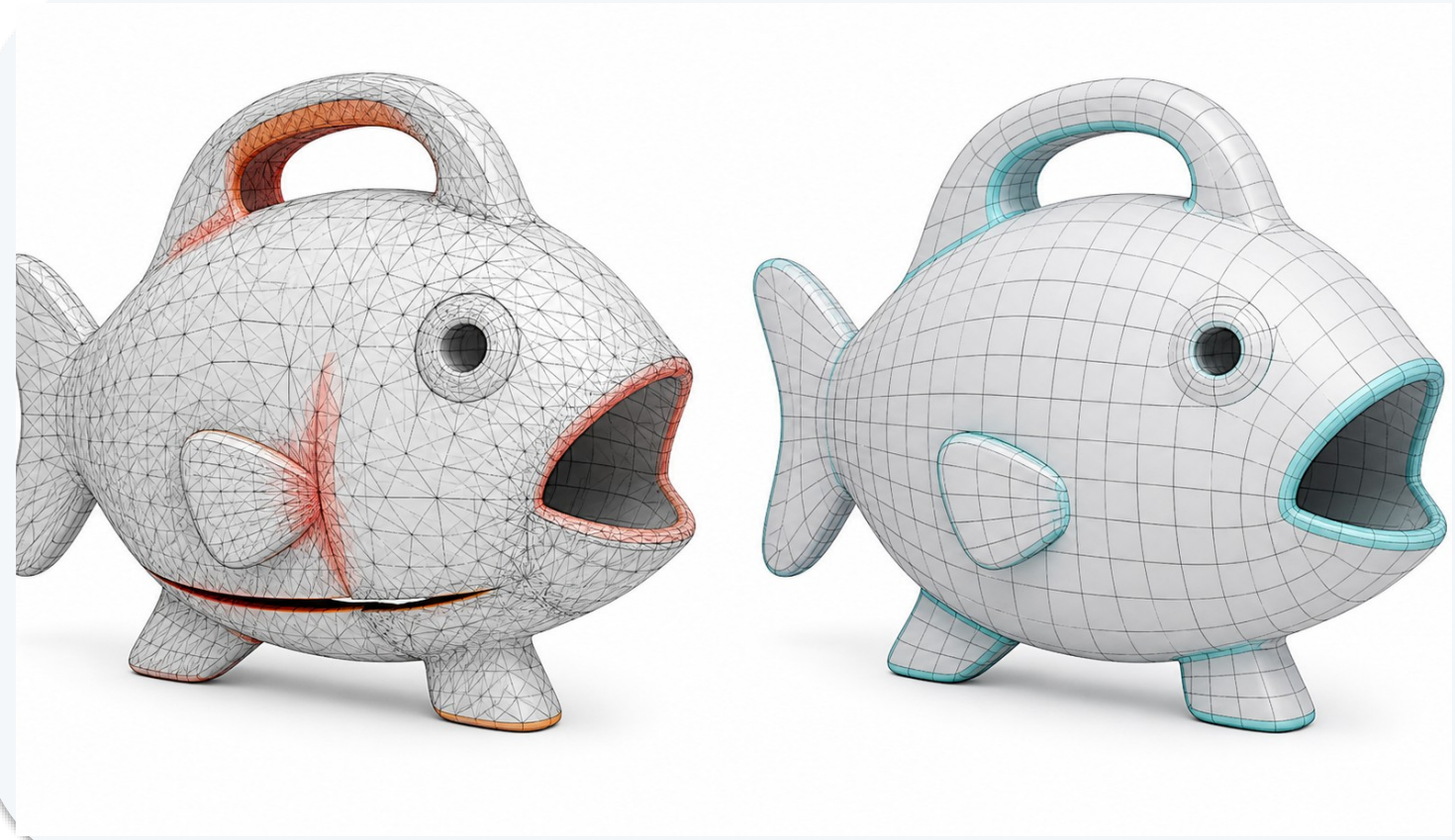
**K4 · K5 · A3**

OBJECTIVE

Measure geometry, transforms, topology and evidence before making repairs.


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# Activity 5 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 5 — quality gates

## Form

- Baseline preserved
- Units and transforms recorded

## Structure

- Topology statistics captured
- Non-manifold and normals checks completed

## Delivery

- Findings have severity and evidence
- A frozen baseline Blender file, defect register and validation dashboard.


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# Activity 5 — acceptance evidence

## ACCEPTANCE CHECKS

- Baseline preserved
- Units and transforms recorded
- Topology statistics captured
- Non-manifold and normals checks completed
- Findings have severity and evidence

## EVIDENCE TO SUBMIT

- Baseline metadata
- Statistics screenshot
- Face Orientation view
- Non-manifold selection
- Defect register


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# Repair Non-Manifold Geometry, Normals and Duplicate Data

ACTIVITY 6

**The inspected model has a belly seam, flipped faces and doubled vertices that cause shading and export defects.**

DELIVERABLE

**A repaired manifold model with before/after evidence and regression results.**

ALIGNMENT

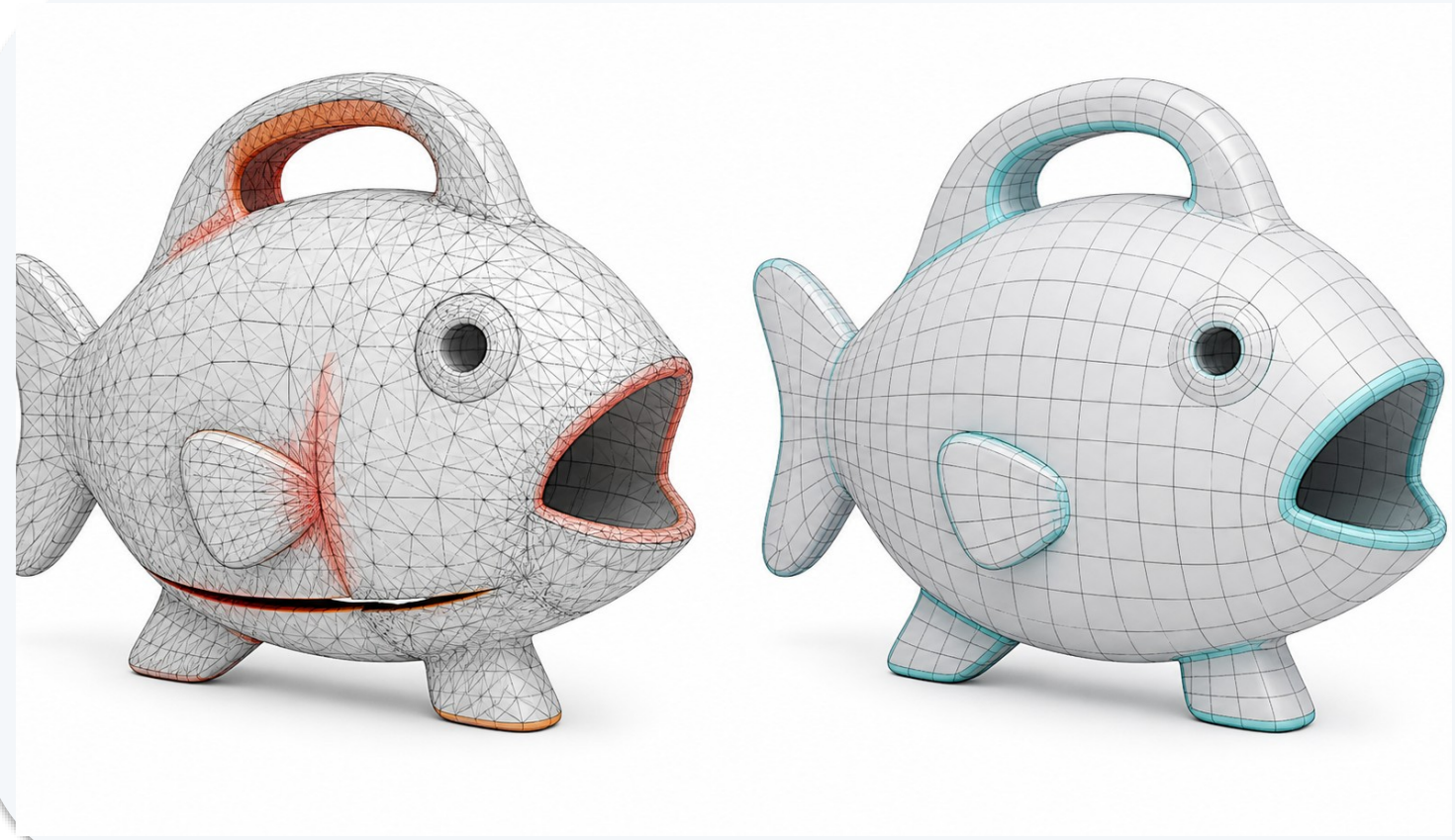
**K4 · A2 · A3**

OBJECTIVE

Apply local, reversible repairs and prove that the original failing checks now pass.


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# Activity 6 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 6 — quality gates

## Form

- Baseline remains unchanged
- Merge threshold documented

## Structure

- Boundary repaired locally
- Normals face outward

## Delivery

- Regression checks pass
- A repaired manifold model with before/after evidence and regression results.


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# Activity 6 — acceptance evidence

## ACCEPTANCE CHECKS

- Baseline remains unchanged
- Merge threshold documented
- Boundary repaired locally
- Normals face outward
- Regression checks pass

## EVIDENCE TO SUBMIT

- Before/after Face Orientation
- Merged-vertex count
- Repaired boundary close-up
- Regression table
- Versioned Blender file


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# Retopologize and Optimize an AI Mesh for Its Target Use

ACTIVITY 7

**A dense generated mesh must become a responsive real-time asset without losing its recognizable silhouette.**

DELIVERABLE

**A target-use retopology, statistics comparison and feature-retention evidence.**

ALIGNMENT

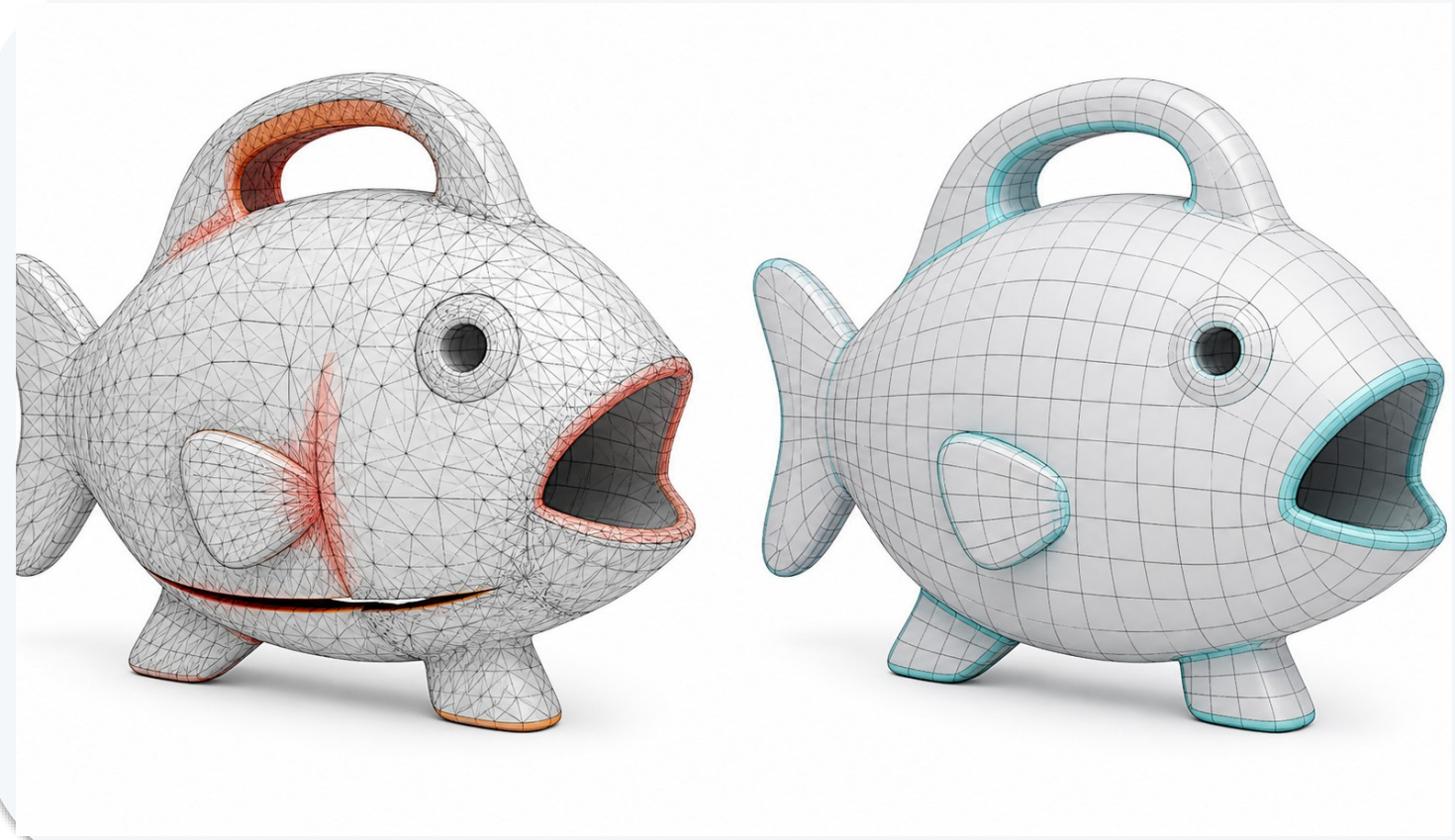
**K2 · K4 · A3 · A4**

OBJECTIVE

Select a retopology approach, meet a polygon budget and preserve critical form.


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# Activity 7 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 7 — quality gates

## Form

- Target use and budget stated
- Method justified

## Structure

- Silhouette retained
- Topology checks pass

## Delivery

- Reduction metrics recorded
- A target-use retopology, statistics comparison and feature-retention evidence.


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# Activity 7 — acceptance evidence

## ACCEPTANCE CHECKS

- Target use and budget stated
- Method justified
- Silhouette retained
- Topology checks pass
- Reduction metrics recorded

## EVIDENCE TO SUBMIT

- High/low overlay
- Wireframe comparison
- Statistics table
- Triangulated preview
- Limitations note


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# UV-Unwrap and Apply AI-Assisted PBR Materials

ACTIVITY 8

**The optimized asset needs a consistent product finish and texture set suitable for review and export.**

DELIVERABLE

**A UV-mapped Blender asset, validated material channels and texture evidence.**

ALIGNMENT

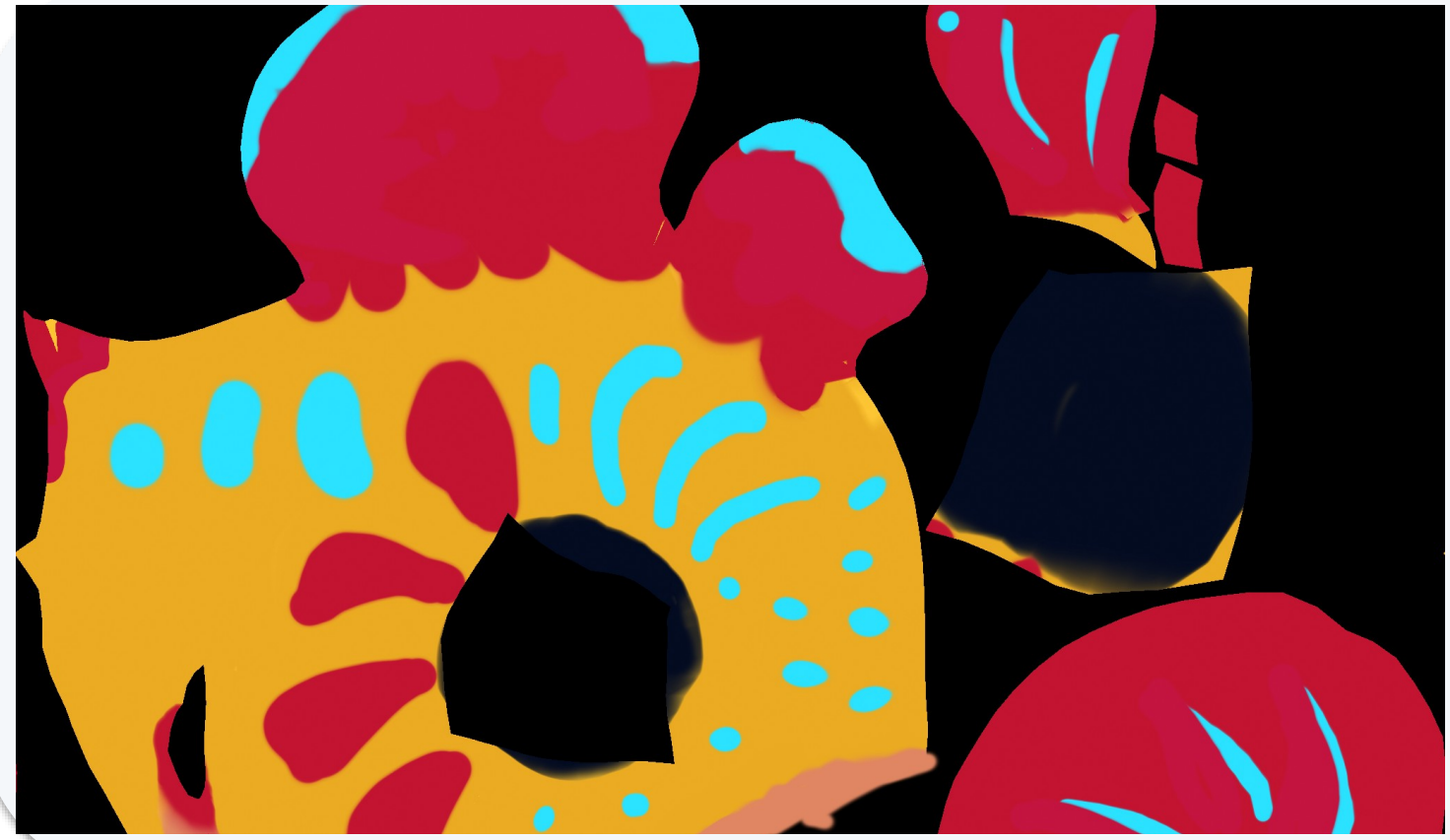
**K2 · K5 · A3 · A4**

OBJECTIVE

Create usable UVs and physically plausible materials from AI-assisted references.


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# Activity 8 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 8 — quality gates

## Form

- Seams follow logical boundaries
- Checker shows acceptable distortion

## Structure

- Texel density is consistent
- Colour spaces are correct

## Delivery

- Texture provenance recorded
- A UV-mapped Blender asset, validated material channels and texture evidence.


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# Activity 8 — acceptance evidence

## ACCEPTANCE CHECKS

- Seams follow logical boundaries
- Checker shows acceptable distortion
- Texel density is consistent
- Colour spaces are correct
- Texture provenance recorded

## EVIDENCE TO SUBMIT

- UV layout screenshot
- Checker test
- Shader graph
- Neutral-light material render
- Texture register


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WHAT YOU CAN NOW DEMONSTRATE

# Topic 2 recap

1

## Concepts

15 decision-ready frameworks

2

## Activities

Activities 5–8 completed

3

## Evidence

Editable Blender files, metrics and presentations

4

## Alignment

K1 · K4 · K5 · A2 · A3 · A4


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TOPIC 3

03

# 3D Model Optimization, Standards and Production Workflows

Standards · rendering · export · interoperability · documentation · capstone delivery


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PROGRESSIVE LEARNING

# Topic 3 — capability map

1

## Understand

Standards

2

## Apply

rendering

3

## Review

K3 · K4 · K5 · K6 · A3 · A5 · A6

4

## Demonstrate

Activities 9–12


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# Target-use matrix

1

## Games

Real-time triangle, texture and draw-call budgets.

2

## Printing

Watertight volume, wall thickness and physical scale.

3

## Animation

Deformation topology, pivots and naming.

4

## Visualisation

High-quality shading, camera and colour management.


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# Decisions and risks — Target-use matrix

## DECISIONS THAT IMPROVE QUALITY

- Choose the target before optimisation.
- Create delivery variants from a controlled master.
- Do not assume one export satisfies every use.
- Test in the destination.

## FAILURE MODES TO CONTROL

- Optimising for one target damages another.
- A beautiful render model may not be printable.
- Format conversion loses data.


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# Target-use matrix

PRINCIPLE

**The same source model needs different preparation for games, 3D printing, animation and visualisation.**

APPLIED EXAMPLE

The approved master produces glTF for web, STL for print validation and a high-resolution Blender render file.

EVIDENCE TO TRACK

1. Target requirements met
2. Variants delivered
3. Round-trip tests


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# Technical conventions for 3D assets

1

## Naming

Role-based object and material names.

2

## Units

Declared measurement system and scale.

3

## Axes

Orientation and forward/up convention.

4

## Hierarchy

Parenting and collection structure.


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# Decisions and risks — Technical conventions for 3D assets

## DECISIONS THAT IMPROVE QUALITY

- Publish a project convention sheet.
- Validate naming automatically where possible.
- Use origin and pivot rules by asset type.
- Record coordinate conversion.

## FAILURE MODES TO CONTROL

- Generic names break automation.
- Mixed units produce assembly failure.
- Unexpected hierarchy changes motion.


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# Technical conventions for 3D assets

PRINCIPLE

**Consistent naming, units, axes, origins, hierarchy and metadata act as the conventions that make digital models understandable across teams.**

APPLIED EXAMPLE

PRD_Speaker_BODY, PRD_Speaker_KNOB and MAT_Plastic_White follow a shared convention and export consistently.

EVIDENCE TO TRACK

1. Naming compliance
2. Unit declaration
3. Hierarchy errors


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# Dimension and tolerance evidence

1

## Nominal

Target size from the brief.

2

## Actual

Measured Blender dimension.

3

## Tolerance

Permissible variation.

4

## Clearance

Required gap between parts.


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# Decisions and risks — Dimension and tolerance evidence

## DECISIONS THAT IMPROVE QUALITY

- Measure critical-to-function features first.
- State whether modifiers affect measured output.
- Use appropriate precision.
- Flag assumptions not supported by source data.

## FAILURE MODES TO CONTROL

- Viewport grids are not measurements.
- Unapplied scale misleads dimension readouts.
- False precision suggests unsupported accuracy.


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# Dimension and tolerance evidence

PRINCIPLE

**Visual models become accountable design artefacts when key dimensions, clearances and tolerances are measured and documented.**

APPLIED EXAMPLE

A knob diameter is 32.0 mm ±0.5 mm and its body clearance is at least 1.0 mm after applied modifiers.

EVIDENCE TO TRACK

1. Critical dimensions within tolerance
2. Clearance minimum
3. Assumptions open


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# Render engines and delivery intent

1

## Workbench

Fast topology and form review.

2

## Eevee

Real-time look development and previews.

3

## Cycles

Path-traced lighting and material realism.

4

## Viewport

Immediate inspection, not final evidence by default.


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# Decisions and risks — Render engines and delivery intent

## DECISIONS THAT IMPROVE QUALITY

- Match engine to decision purpose.
- Use Workbench overlays for mesh evidence.
- Control samples and denoising.
- Record engine and colour settings.

## FAILURE MODES TO CONTROL

- Engine differences change material appearance.
- Denoising can smear detail.
- Uncontrolled colour management changes approval renders.


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# Render engines and delivery intent

PRINCIPLE

**Choose Eevee for speed and interactive review, Cycles for physically richer final rendering, and Workbench for structural evidence.**

APPLIED EXAMPLE

A topology review uses Workbench wireframe; the design review uses Eevee; the final marketing still uses Cycles.

EVIDENCE TO TRACK

1. Render time
2. Noise level
3. Settings recorded


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# Camera and composition

1

## Focal length

Controls perspective compression.

2

## Framing

Shows whole form with intentional margins.

3

## Angle

Reveals relevant features.

4

## Consistency

Locks comparison across variants.


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# Decisions and risks — Camera and composition

## DECISIONS THAT IMPROVE QUALITY

- Use orthographic views for dimensions.
- Use moderate focal lengths for product perspective.
- Lock camera after approval.
- Render required evidence views systematically.

## FAILURE MODES TO CONTROL

- Wide-angle distortion exaggerates form.
- Auto framing changes comparison.
- Important features are occluded.


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# Camera and composition

PRINCIPLE

**A consistent camera turns renders into comparable evidence and communicates scale, function and hierarchy.**

APPLIED EXAMPLE

A 70 mm three-quarter camera is duplicated across variants so differences come from design, not perspective.

EVIDENCE TO TRACK

1. Required views complete
2. Camera parity
3. Margin consistency


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# Colour management

1

## Exposure

Overall scene brightness.

2

## View transform

Maps dynamic range to display.

3

## Look

Optional contrast treatment.

4

## Output

Colour space and file format.


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# Decisions and risks — Colour management

## DECISIONS THAT IMPROVE QUALITY

- Lock settings before variant comparison.
- Avoid editing material values to compensate for exposure.
- Use lossless output for evidence masters.
- Document display limitations.

## FAILURE MODES TO CONTROL

- Different displays still vary.
- Embedded profiles may be stripped.
- Clipping hides material differences.


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# Colour management

PRINCIPLE

**Colour management maps scene-linear values to the display; consistent settings are essential for fair review and predictable output.**

APPLIED EXAMPLE

All approval renders use the same AgX-based view settings, exposure and studio lights.

EVIDENCE TO TRACK

1. Exposure locked
2. Output profile
3. Clipped pixels


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# Export formats

1

## glTF/GLB

Efficient real-time delivery with PBR materials.

2

## FBX

Broad DCC/engine interchange, but settings vary.

3

## OBJ

Simple geometry and UV interchange.

4

## STL

Triangle surface for printing; little scene metadata.


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# Decisions and risks — Export formats

## DECISIONS THAT IMPROVE QUALITY

- Choose by required data, not familiarity.
- Apply target axis and scale settings.
- Embed or package textures deliberately.
- Re-import to verify.

## FAILURE MODES TO CONTROL

- Materials are lost or remapped.
- Scale changes silently.
- Modifiers or instances export unexpectedly.


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# Export formats

PRINCIPLE

**File formats preserve different subsets of geometry, materials, hierarchy, animation and units.**

APPLIED EXAMPLE

A web asset ships as GLB with packed PBR textures; a print test ships as millimetre-scale STL after manifold validation.

EVIDENCE TO TRACK

1. Round-trip pass
2. File size
3. Required data retained


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# Game-ready optimization

1

## Geometry

Triangle count and LODs.

2

## Materials

Minimise slots and draw calls.

3

## Textures

Resolution, compression and atlasing.

4

## Collision

Simple collision representation.


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# Decisions and risks — Game-ready optimization

## DECISIONS THAT IMPROVE QUALITY

- Triangulate predictably before final check.
- Bake high-detail normals onto low geometry.
- Use consistent texel density.
- Test inside the target engine.

## FAILURE MODES TO CONTROL

- Hidden geometry wastes budget.
- Too many material slots increase draw calls.
- Normal-map seams appear after triangulation.


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# Game-ready optimization

PRINCIPLE

**Real-time assets balance silhouette, shading, textures and draw calls against a defined platform budget.**

APPLIED EXAMPLE

The hero asset uses one 2K texture set and three LODs; a simple convex collision mesh is delivered separately.

EVIDENCE TO TRACK

1. Triangles
2. Draw calls
3. Texture memory


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# 3D-print preparation

1

## Volume

Closed and manifold.

2

## Thickness

Meets process/material limits.

3

## Overhang

Compatible with orientation and supports.

4

## Tolerance

Allows fit after process variation.


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# Decisions and risks — 3D-print preparation

## DECISIONS THAT IMPROVE QUALITY

- Set units to millimetres deliberately.
- Check thin walls and disconnected shells.
- Orient to reduce support and weak layers.
- Use a slicer as a second validation environment.

## FAILURE MODES TO CONTROL

- Surface-only models produce no solid.
- Tiny gaps create slicing artefacts.
- Tight clearances fuse moving parts.


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# 3D-print preparation

PRINCIPLE

**Print preparation turns a visible surface into a manufacturable volume with scale, thickness, orientation and support considerations.**

APPLIED EXAMPLE

A hollow product shell gains 2.0 mm wall thickness, drain holes and a flat print orientation before STL export.

EVIDENCE TO TRACK

1. Wall thickness
2. Estimated support
3. Slicer warnings


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# Animation-ready organization

1

## Pivots

Rotate around meaningful mechanical centres.

2

## Hierarchy

Parent-child motion relationships.

3

## Topology

Supports deformation where needed.

4

## Naming

Enables rigging and export automation.


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# Decisions and risks — Animation-ready organization

## DECISIONS THAT IMPROVE QUALITY

- Test one representative motion.
- Apply scale before rigging where appropriate.
- Keep moving parts separate.
- Avoid modifiers unsupported by the target.

## FAILURE MODES TO CONTROL

- Wrong pivot makes motion orbit.
- Applied transforms after rigging can break animation.
- Merged moving parts cannot be controlled independently.


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# Animation-ready organization

PRINCIPLE

**Animation depends on predictable pivots, transforms, topology, hierarchy and deformation—not merely a visually complete mesh.**

APPLIED EXAMPLE

A control knob remains a separate object with its origin on the rotational axis and a clear parent relationship.

EVIDENCE TO TRACK

1. Motion test passed
2. Pivot alignment
3. Hierarchy compliance


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# Documentation package

1

## Source

Clean Blender master with organised data.

2

## Exports

Target-specific files and textures.

3

## Preview

Consistent renders or turntable.

4

## Record

Brief, changes, checks and known limitations.


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# Decisions and risks — Documentation package

## DECISIONS THAT IMPROVE QUALITY

- Use a predictable folder structure.
- Include version and date.
- Separate final from working files.
- State open risks explicitly.

## FAILURE MODES TO CONTROL

- Missing texture dependencies.
- Final files mixed with drafts.
- No note on target version or units.


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# Documentation package

PRINCIPLE

**A production-ready handoff includes the editable source, exports, previews, specifications, provenance and validation evidence.**

APPLIED EXAMPLE

The handoff contains SOURCE, EXPORT, TEXTURES, PREVIEW and EVIDENCE folders plus a readme and acceptance checklist.

EVIDENCE TO TRACK

1. Package completeness
2. Broken links
3. Open limitations


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# Automated checks

1

## Rules

Machine-checkable requirements.

2

## Report

Object-level findings with severity.

3

## Repair

Optional and narrowly scoped.

4

## Review

Human confirmation of geometry and intent.


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# Decisions and risks — Automated checks

## DECISIONS THAT IMPROVE QUALITY

- Run checks without silent mutation.
- Show exactly what failed.
- Allow thresholds to be configured.
- Version the report with the asset.

## FAILURE MODES TO CONTROL

- False confidence from incomplete checks.
- Auto-fixes can damage valid exceptions.
- Tool version changes break scripts.


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# Automated checks

PRINCIPLE

**Small scripts can standardise repetitive checks such as naming, transforms, dimensions and object counts, while human review remains essential.**

APPLIED EXAMPLE

A validation script reports unnamed objects, non-unit scale and missing UV maps; it does not automatically modify them.

EVIDENCE TO TRACK

1. Rules passed
2. False positives
3. Checks with evidence


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# Stakeholder design review

1

## Context

Problem and intended use.

2

## Options

Comparable design variants.

3

## Evidence

Metrics and validation views.

4

## Decision

Recommendation, trade-offs and next actions.


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# Decisions and risks — Stakeholder design review

## DECISIONS THAT IMPROVE QUALITY

- Lead with the decision required.
- Show comparable views.
- Distinguish concept from verified evidence.
- Make unresolved risks visible.

## FAILURE MODES TO CONTROL

- Beautiful renders crowd out technical evidence.
- Unweighted criteria obscure trade-offs.
- Stakeholders cannot trace the approved file.


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# Stakeholder design review

PRINCIPLE

**A review presentation should connect requirements, options, evidence, risks and recommendation in a decision-ready story.**

APPLIED EXAMPLE

The review recommends Variant B because it passes all functional criteria with a modest polygon increase and lower manufacturing risk.

EVIDENCE TO TRACK

1. Decision captured
2. Evidence linked
3. Actions assigned


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# Organisational alignment

1

## Audience

User needs and accessibility.

2

## Brand

Form language and visual consistency.

3

## Operations

Tools, skills and production constraints.

4

## Lifecycle

Reuse, maintenance and future variants.


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# Decisions and risks — Organisational alignment

## DECISIONS THAT IMPROVE QUALITY

- Evaluate business fit alongside technical quality.
- Prefer reusable modular structures.
- Capture ownership and maintenance.
- Reject novelty that increases risk without value.

## FAILURE MODES TO CONTROL

- A technically perfect asset may not fit the brand.
- Vendor-specific features reduce portability.
- No owner exists for future updates.


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# Organisational alignment

PRINCIPLE

**An asset succeeds when it supports the organisation's audience, brand, production capacity, budget, risk appetite and reuse strategy.**

APPLIED EXAMPLE

A modular product model enables colour variants without duplicating geometry, reducing future campaign effort.

EVIDENCE TO TRACK

1. Reuse potential
2. Production effort
3. Strategic-fit score


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# Final acceptance and release

1

## Verify

Run the full checklist on the release candidate.

2

## Approve

Record competent reviewer sign-off.

3

## Package

Freeze source, exports and evidence.

4

## Archive

Retain prior versions and provenance.


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# Decisions and risks — Final acceptance and release

## DECISIONS THAT IMPROVE QUALITY

- Never approve a moving target.
- Re-run checks after the last change.
- Use checksums or immutable version labels.
- Separate answer keys and trainer-only material from learner publication.

## FAILURE MODES TO CONTROL

- Last-minute edits bypass checks.
- Wrong file is linked to the LMS.
- Confidential assessment keys become public.


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# Final acceptance and release

PRINCIPLE

**Release is a controlled decision based on complete evidence, resolved blockers and explicit acceptance of any remaining limitations.**

APPLIED EXAMPLE

v08 is frozen after round-trip export, render, scale and topology checks; the handoff is signed and v07 archived.

EVIDENCE TO TRACK

1. Release checks passed
2. Approvals
3. Artifact checksum


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# Topic 3 activities

## Activities 9–10

- Create a Neutral Studio Render and Visual QA Set
- Prepare and Verify a Game/Visualization Export

## Activities 11–12

- Prepare and Verify a 3D-Printable STL
- Capstone: Deliver an AI-Assisted 3D Asset Review Package

## Progression

- Brief or inspect
- Build or repair
- Verify and present


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# Create a Neutral Studio Render and Visual QA Set

ACTIVITY 9

**The design team needs evidence renders that reveal form and surface quality rather than hiding defects with dramatic lighting.**

DELIVERABLE

**A four-view neutral render set and a documented render configuration.**

ALIGNMENT

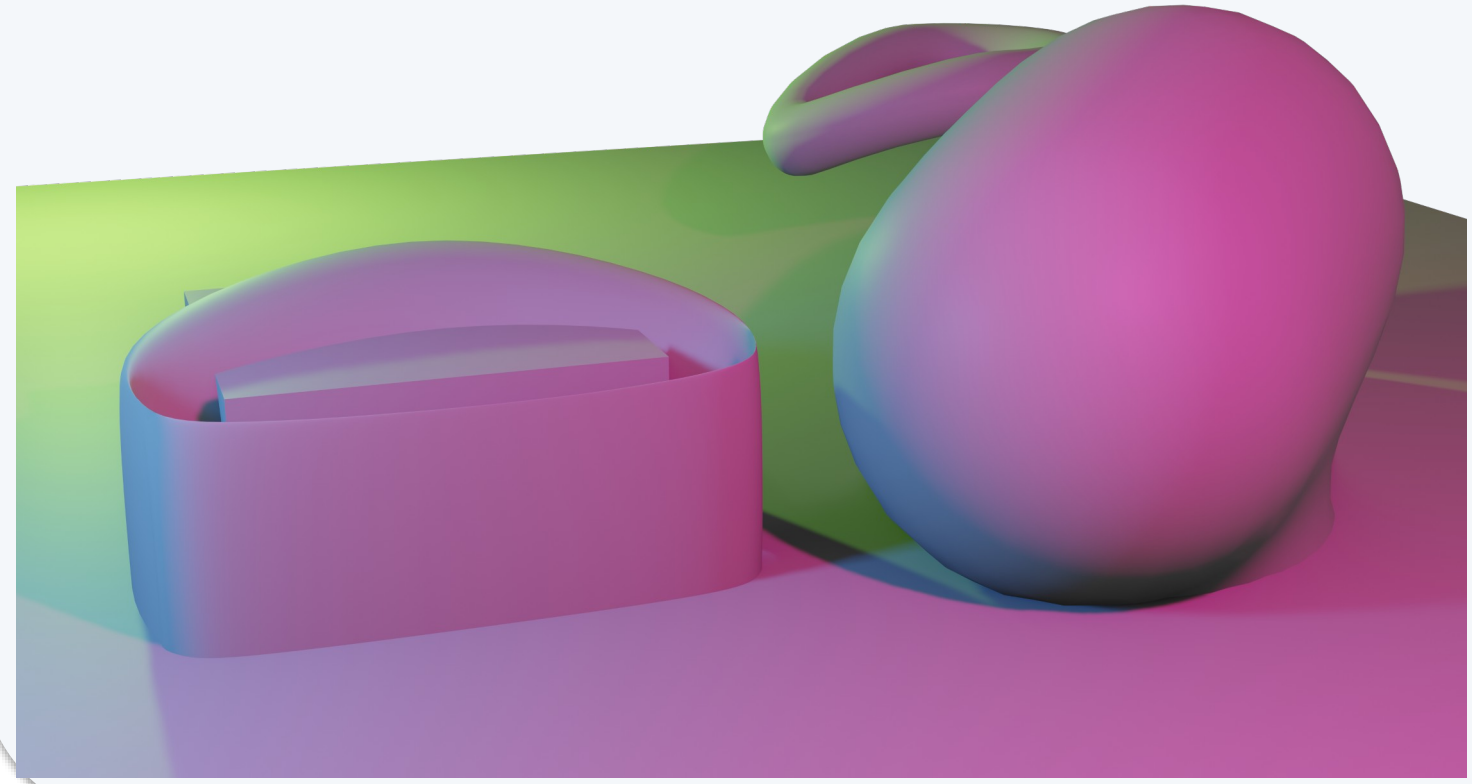
**K3 · K5 · A3**

OBJECTIVE

Use consistent cameras, lighting and colour settings to create comparable validation evidence.


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# Activity 9 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 9 — quality gates

## Form

- Four required views complete
- Camera and exposure consistent

## Structure

- Neutral lighting reveals form
- Render settings recorded

## Delivery

- Blocking artefacts resolved
- A four-view neutral render set and a documented render configuration.


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# Activity 9 — acceptance evidence

## ACCEPTANCE CHECKS

- Four required views complete
- Camera and exposure consistent
- Neutral lighting reveals form
- Render settings recorded
- Blocking artefacts resolved

## EVIDENCE TO SUBMIT

- Camera setup
- Lighting diagram
- Four-view render set
- Settings table
- Surface QA note


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# Prepare and Verify a Game/Visualization Export

ACTIVITY 10

**A client needs a GLB asset for a web product viewer with predictable scale, materials and performance.**

DELIVERABLE

**A validated GLB export, round-trip Blender check and delivery record.**

ALIGNMENT

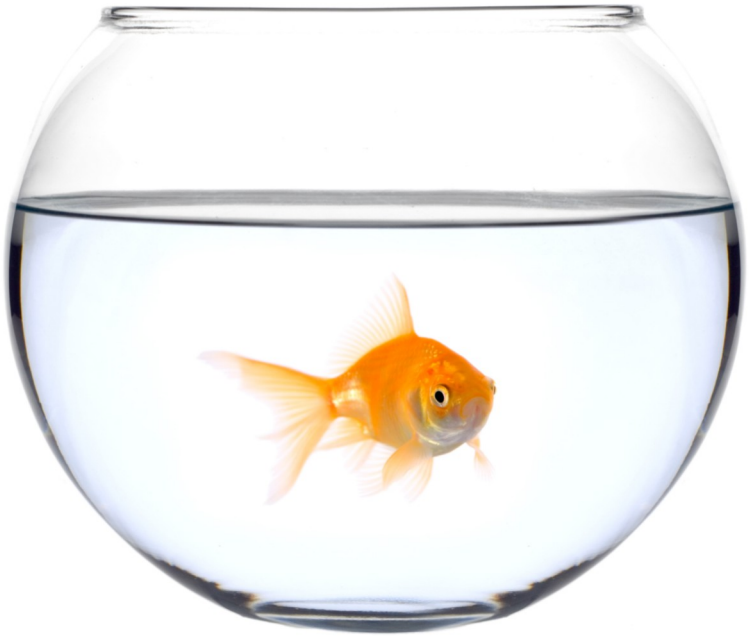
**K3 · K4 · K6 · A4 · A6**

OBJECTIVE

Package a model for real-time delivery and verify it through round-trip import.


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# Activity 10 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 10 — quality gates

## Form

- Target requirements stated
- Only approved objects exported

## Structure

- Scale and axes correct
- Materials survive round trip

## Delivery

- Triangle budget met
- A validated GLB export, round-trip Blender check and delivery record.


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# Activity 10 — acceptance evidence

## ACCEPTANCE CHECKS

- Target requirements stated
- Only approved objects exported
- Scale and axes correct
- Materials survive round trip
- Triangle budget met

## EVIDENCE TO SUBMIT

- Export settings
- Final statistics
- Source/import comparison
- Target-viewer screenshot
- Delivery package


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# Prepare and Verify a 3D-Printable STL

ACTIVITY 11

**The product concept must be delivered as a small physical prototype for ergonomic review.**

DELIVERABLE

**A millimetre-scale STL, printable-source Blender file and slicer-readiness checklist.**

ALIGNMENT

**K4 · K5 · A4 · A6**

OBJECTIVE

Convert a model into a watertight, correctly scaled print candidate with manufacturability evidence.


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# Activity 11 — example evidence




**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 11 — quality gates

## Form

- Print process and limits stated
- Closed manifold volume

## Structure

- Wall thickness meets target
- Re-import scale correct

## Delivery

- Slicer warnings resolved or documented
- A millimetre-scale STL, printable-source Blender file and slicer-readiness checklist.


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# Activity 11 — acceptance evidence

## ACCEPTANCE CHECKS

- Print process and limits stated
- Closed manifold volume
- Wall thickness meets target
- Re-import scale correct
- Slicer warnings resolved or documented

## EVIDENCE TO SUBMIT

- Dimension screenshot
- Manifold/Face Orientation checks
- Wall-thickness evidence
- Re-import comparison
- Slicer or readiness report


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# Capstone: Deliver an AI-Assisted 3D Asset Review Package

ACTIVITY 12

**You are accountable for a complete product-asset handoff that will be reviewed by design, production and business stakeholders.**

DELIVERABLE

**A release-candidate Blender asset, target export, validation dashboard and decision-ready presentation.**

ALIGNMENT

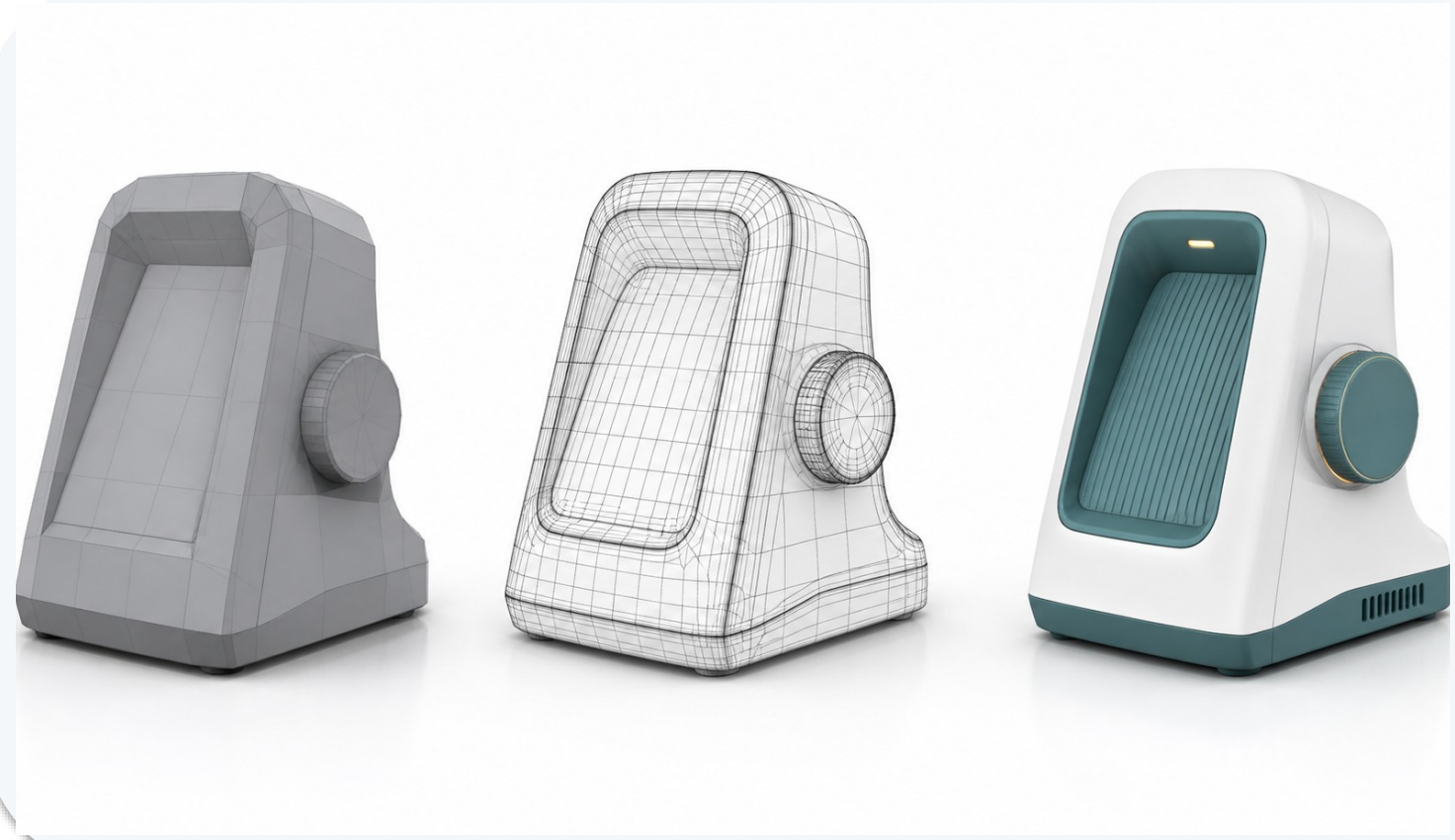
**K1–K6 · A1–A6**

OBJECTIVE

Integrate brief, AI assistance, Blender refinement, validation, export and stakeholder recommendation.


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# Activity 12 — example evidence



**Use this as a visual benchmark, then replace it with your own Blender view. Your submission must show the editable source, measured checks and acceptance result—not only a polished render.**

Concept visual — validate against the editable Blender file.


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ACCEPTANCE CRITERIA

# Activity 12 — quality gates

## Form

- All K/A areas evidenced
- Baseline and versions traceable

## Structure

- Blocking checks passed
- Round-trip export verified

## Delivery

- Recommendation and limitations clear
- A release-candidate Blender asset, target export, validation dashboard and decision-ready presentation.


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# Activity 12 — acceptance evidence

## ACCEPTANCE CHECKS

- All K/A areas evidenced
- Baseline and versions traceable
- Blocking checks passed
- Round-trip export verified
- Recommendation and limitations clear

## EVIDENCE TO SUBMIT

- Brief and provenance
- Version history
- Validation dashboard
- Final render set
- Round-trip result
- Stakeholder presentation


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WHAT YOU CAN NOW DEMONSTRATE

# Topic 3 recap

1

## Concepts

15 decision-ready frameworks

2

## Activities

Activities 9–12 completed

3

## Evidence

Editable Blender files, metrics and presentations

4

## Alignment

K3 · K4 · K5 · K6 · A3 · A5 · A6


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COURSE WRAP-UP

04

# Review, assessment and release

From AI-assisted idea to accountable 3D asset


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LEARNING OUTCOMES

# What you achieved

1

## Generate

Translate requirements into prompts, scripts and controlled variants.

2

## Refine

Repair topology, normals, UVs and materials in Blender.

3

## Validate

Use metrics and evidence rather than appearance alone.

4

## Deliver

Prepare target-specific renders, exports and documentation.


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END-TO-END CAPABILITY

# Production workflow

1. Brief and acceptance criteria
2. AI-assisted concept or script
3. Blender refinement
4. Validation and target export
5. Release package and approval


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WHY IT MATTERS

# The accountable workflow

**AI accelerates possibilities; Blender and human review turn them into production evidence.**

Release only what you can explain, validate and reproduce.


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DOWNLOAD AND SUBMIT

# Courseware and assessment on the LMS

## Courseware

- Trainer slides
- Learner slides
- Learner Guide and Lesson Plan

## Activities

- Instructions and checklists
- Starter/example Blender files
- Sample evidence presentations

## Assessment

- Complete the WA then PP
- Upload question papers only
- Answer keys remain trainer-only

**https://lms-tms.tertiaryinfotech.com/**


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DAY 4 · 3:00 PM–6:00 PM

# Assessment reminder

## WA

- 6 SAQs · K1–K6
- 1 hour
- Open book

## PP

- 3 Blender tasks · A1–A6
- 2 hours
- Open book

## Complete

- Assessment attendance
- Upload to LMS
- Sign summary record


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FINAL REMINDER

# Assessment flow

1. TRAQOM / SSG attendance
2. Written Assessment (SAQ)
3. Practical Performance (PP)
4. Upload answers to LMS
5. Sign Assessment Summary Record


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# Digital Attendance (Mandatory)

1

## Scan

Use the QR code shown by the trainer or administrator.

2

## Submit

Confirm the assessment attendance is recorded.

3

## Complete

Sit the WA and PP within the declared timing.

4

## Retain

Follow trainer instructions for assessment records.


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WHY IT MATTERS

# Thank you

**You are ready to build, validate and deliver AI-assisted 3D assets with Blender.**

Create boldly. Validate rigorously.


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